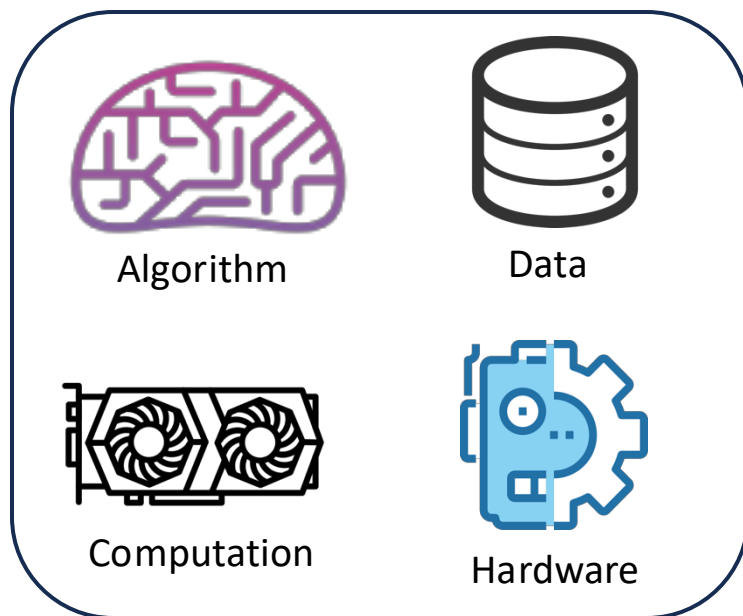


Robot Learning: What, Why, and How?



Embodied intelligence in the physical world



Instructor: Guanya Shi

Recording Reminder

- ☐ This session will be audio recorded for educational use by other students in this course.

Logistics

☐ OH starts from next week

☐ Any questions?

Today's Agenda

Last lecture:

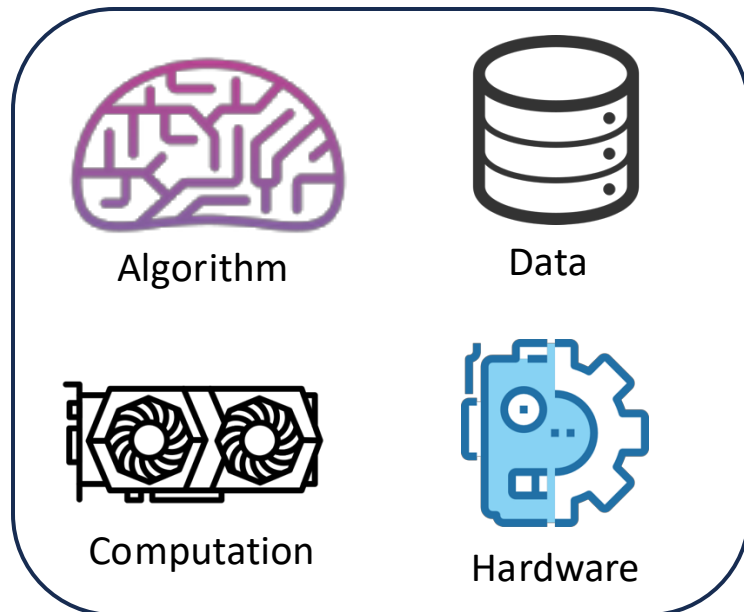
- ☐ What is robot learning and what is its uniqueness?
- ☐ What is the goal of robot learning?
- ☐ Where are we today? How far are we from the goal?
 - In last lecture, we discussed some non-learning methods

Today:

- ☐ Where are we today? How far are we from the goal?
 - A glimpse of today's robot learning
- ☐ Course structure and course objective
- ☐ Relation to other CMU courses
- ☐ Applications of sequential decision making in other domains

A Glimpse of Today's Robot Learning

- ❑ In this class, we will learn how these methods work and more!
 - Model-free reinforcement learning (RL)
 - Sim2Real and Real2Sim2Real
 - Online adaptation
 - Imitation learning with demonstrations
 - Model-based RL
 - Offline RL



How far are we?

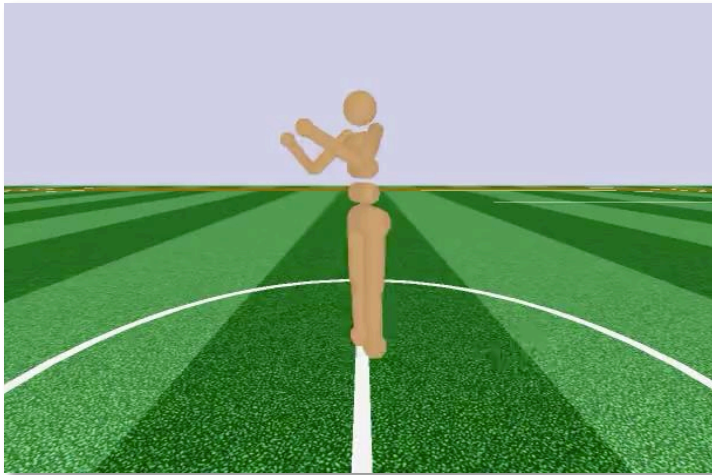


Embodied AGI in the real physical world



Where are We Today: Robot Learning in Simulation

❑ Robot in simulations



[Schulman *et al.*, 2015, 2017]



DM Control Soccer



Article | Published: 09 February 2022

Outracing champion Gran Turismo drivers with deep reinforcement learning

[Peter R. Wurman](#) , [Samuel Barrett](#), [Kenta Kawamoto](#), [James MacGlashan](#), [Kaushik Subramanian](#), [Thomas J. Walsh](#), [Roberto Capobianco](#), [Alisa Devlic](#), [Franziska Eckert](#), [Florian Fuchs](#), [Leilani Gilpin](#), [Piyush Khandelwal](#), [Varun Kompella](#), [HaoChih Lin](#), [Patrick MacAlpine](#), [Declan Oller](#), [Takuma Seno](#), [Craig Sherstan](#), [Michael D. Thomure](#), [Houmehr Aghabozorgi](#), [Leon Barrett](#), [Rory Douglas](#), [Dion Whitehead](#), [Peter Dürr](#), [Peter Stone](#), [Michael Spranger](#) & [Hiroaki Kitano](#) [— Show fewer authors](#)

Where are We Today: Robot Learning in Simulation

- ❑ GT Sophy overtakes the human driver in a fair way (no collision or blocking)
- ❑ How? DRL + distributed training platform + complex reward design



Captured on PS4. Game progression required to access most vehicles.

Where are We Today: Robot Learning in Simulation

- ❑ Eureka: pen spinning
- ❑ How? LLM for reward design

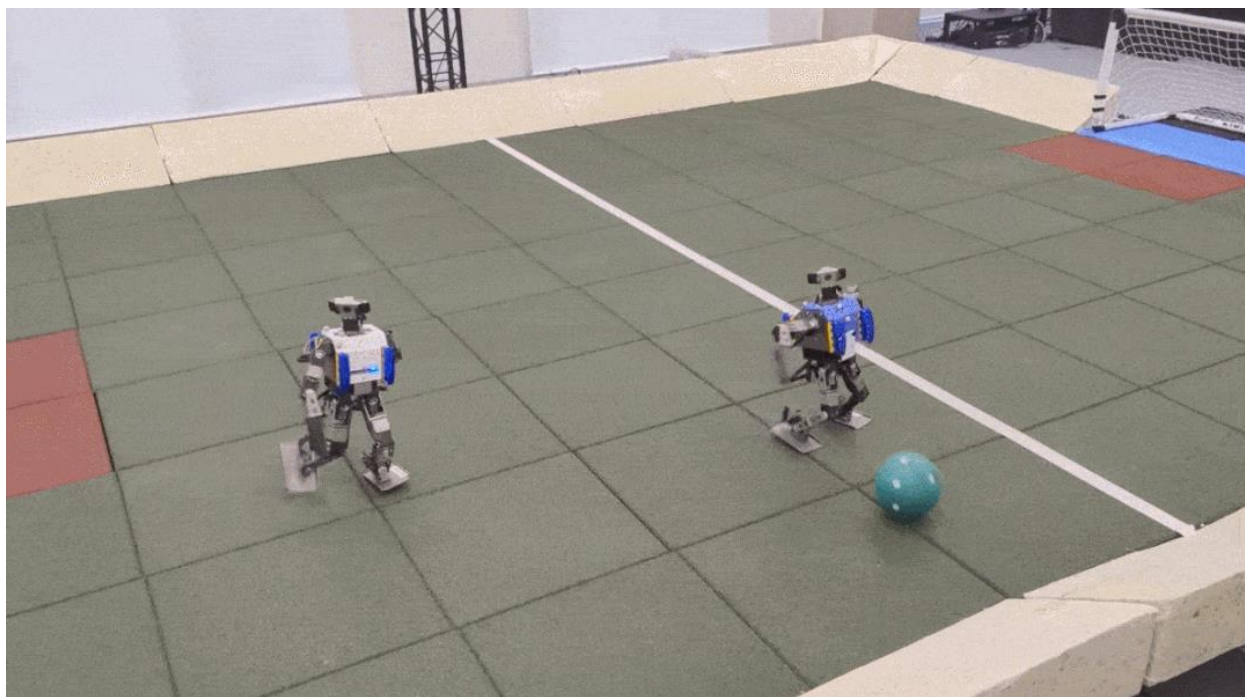
Eureka: Human-Level Reward Design via Coding Large Language Models

Jason Ma ^{1 2}, William Liang², Guanzhi Wang^{1 3}, De-An Huang¹,
Osbert Bastani², Dinesh Jayaraman², Yuke Zhu^{1 4}, Linxi "Jim" Fan ^{1 ‡}, Anima Anandkumar^{1 3 ‡}



Where are We Today: DRL + Sim2Real

- ❑ Train in simulation, deploy in the real world



1-1 Soccer (DeepMind)



2023-4-27

Learning Agile Soccer Skills for a Bipedal Robot with Deep Reinforcement Learning

Tuomas Haarnoja^{*1}, Ben Moran^{*1}, Guy Lever^{*1}, Sandy H. Huang^{*1}, Dhruva Tirumala¹, Markus Wulfmeier¹, Jan Humplik¹, Saran Tunyasuvunakool¹, Noah Y. Siegel¹, Roland Hafner¹, Michael Bloesch¹, Kristian Hartikainen^{2,4}, Arunkumar Byravan¹, Leonard Hasenclever¹, Yuval Tassa¹, Fereshteh Sadeghi^{3,4}, Nathan Batchelor¹, Federico Casarini¹, Stefano Saliceti¹, Charles Game¹, Neil Sreendra, Kushal Patel, Marlon Gwira, Andrea Huber¹, Nicole Hurley¹, Francesco Nori¹, Raia Hadsell¹ and Nicolas Heess¹

^{*}Equal contributions, ¹DeepMind, ²University of Oxford, ³Google, ⁴Work done at DeepMind

Where are We Today: DRL + Sim2Real

- ❑ Train in simulation, deploy in the real world



Agile But Safe: Learning Collision-Free High-Speed Legged Locomotion,
Outstanding Student Paper Award Finalist, RSS'24

[\(https://agile-but-safe.github.io/\)](https://agile-but-safe.github.io/)



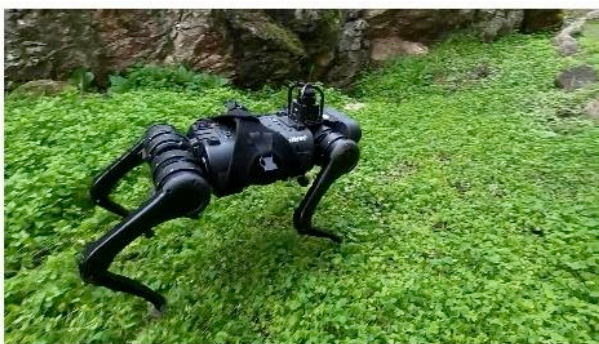
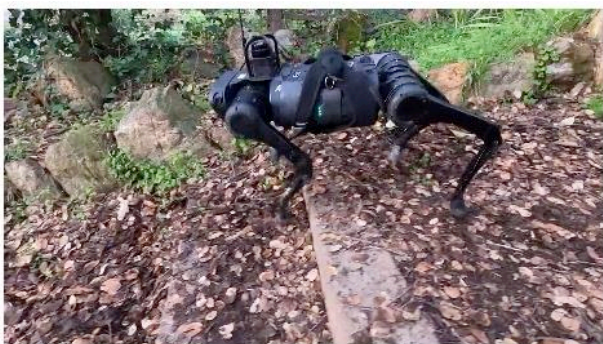
Reinforcement Learning for Versatile, Dynamic, and
Robust Bipedal Locomotion Control

Zhongyu Li¹, Xue Bin Peng², Pieter Abbeel¹, Sergey Levine¹, Glen Berseth^{3,4}, Koushil Sreenath¹

¹University of California Berkeley ²Simon Fraser University ³Université de Montréal ⁴Mila – Quebec AI Institute

Where are We Today: DRL + Sim2Real + Adaptation

- Train in simulation, deploy in the real world with real-time adaptation



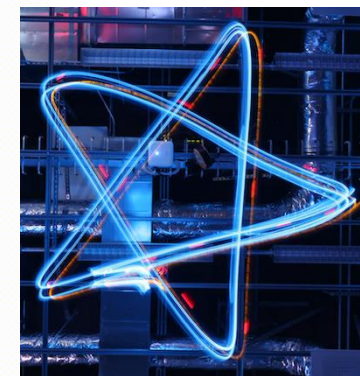
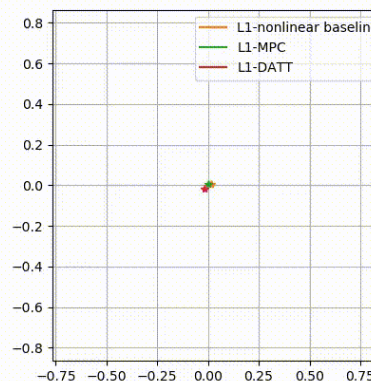
RMA: Rapid Motor Adaptation for Legged Robots

Ashish Kumar
UC Berkeley

Zipeng Fu
Carnegie Mellon University

Deepak Pathak
Carnegie Mellon University

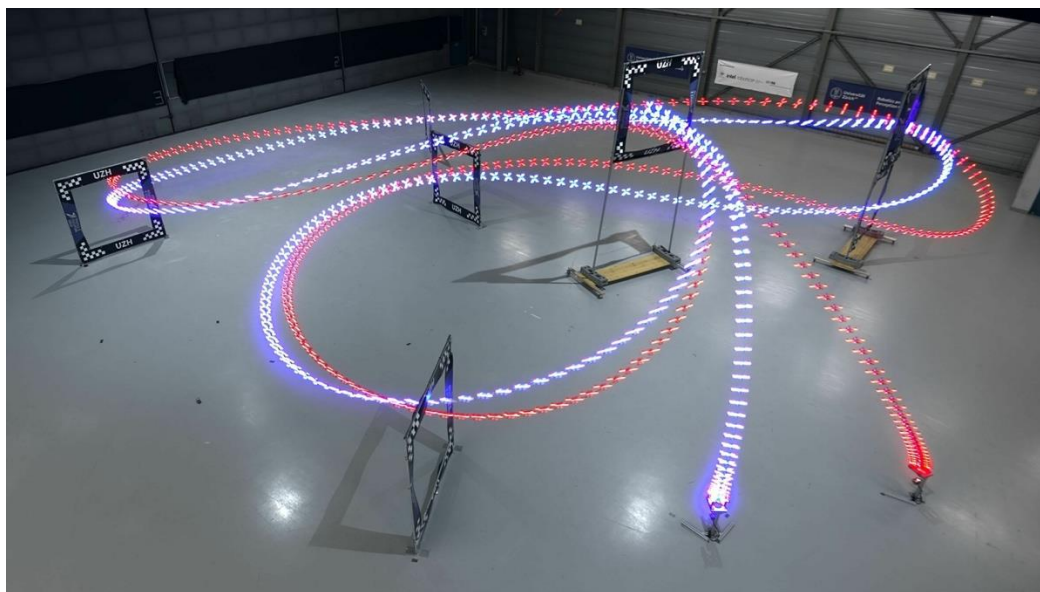
Jitendra Malik
UC Berkeley, Facebook



Huang, Rana, Spitzer, Shi, Boots, DATT: Deep Adaptive Trajectory Tracking, CoRL (oral), 2023

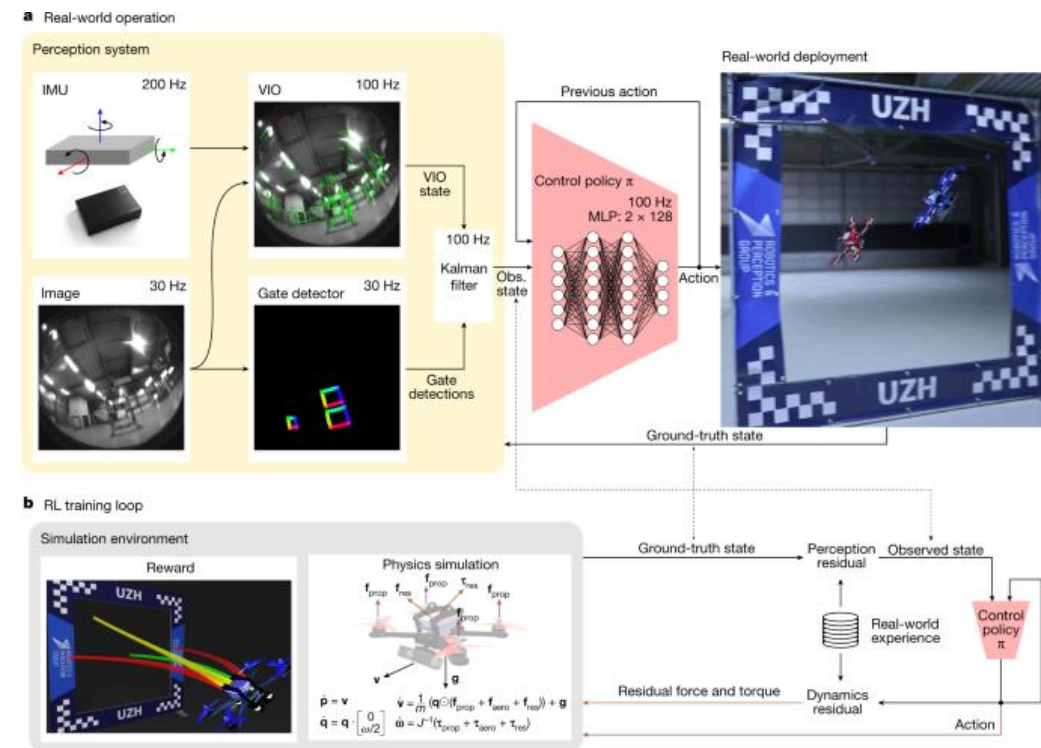
Where are We Today: DRL + Real2Sim2Real

- ❑ Use real data to build/augment/improve a simulator, train DRL in simulation, deploy in the real world



Champion-level drone racing using deep reinforcement learning

[Elia Kaufmann](#) , [Leonard Bauersfeld](#), [Antonio Loquercio](#), [Matthias Müller](#), [Vladlen Koltun](#) & [Davide Scaramuzza](#)



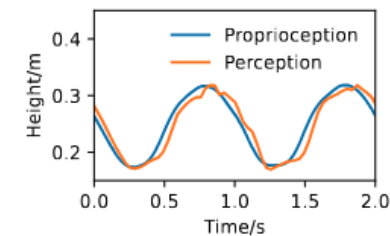
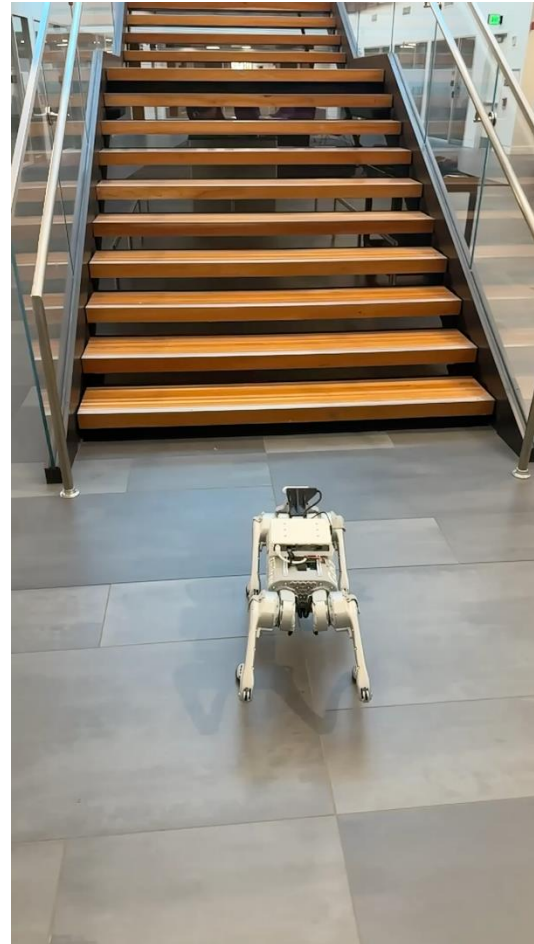
Where are We Today: DRL + Real2Sim2Real

- ❑ Learn a simulator, train DRL in simulation, deploy in the real world

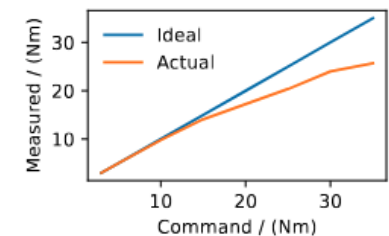


Where are We Today: DRL + Real2Sim2Real

- ❑ Use real data to build/augment/improve a simulator, train DRL in simulation, deploy in the real world



(a) Camera Latency.



(b) Torque Saturation.

Where are We Today: Learning from Demonstrations

- ❑ Collect human demonstrations and do imitation learning

Mobile ALOHA

Learning Bimanual Mobile Manipulation with
Low-Cost Whole-Body Teleoperation

Zipeng Fu* Tony Z. Zhao* Chelsea Finn



Where are We Today: Learning from Demonstrations

- Demonstrations from teleoperation

Mobile ALOHA

Learning Bimanual Mobile Manipulation with
Low-Cost Whole-Body Teleoperation

Zipeng Fu* Tony Z. Zhao* Chelsea Finn

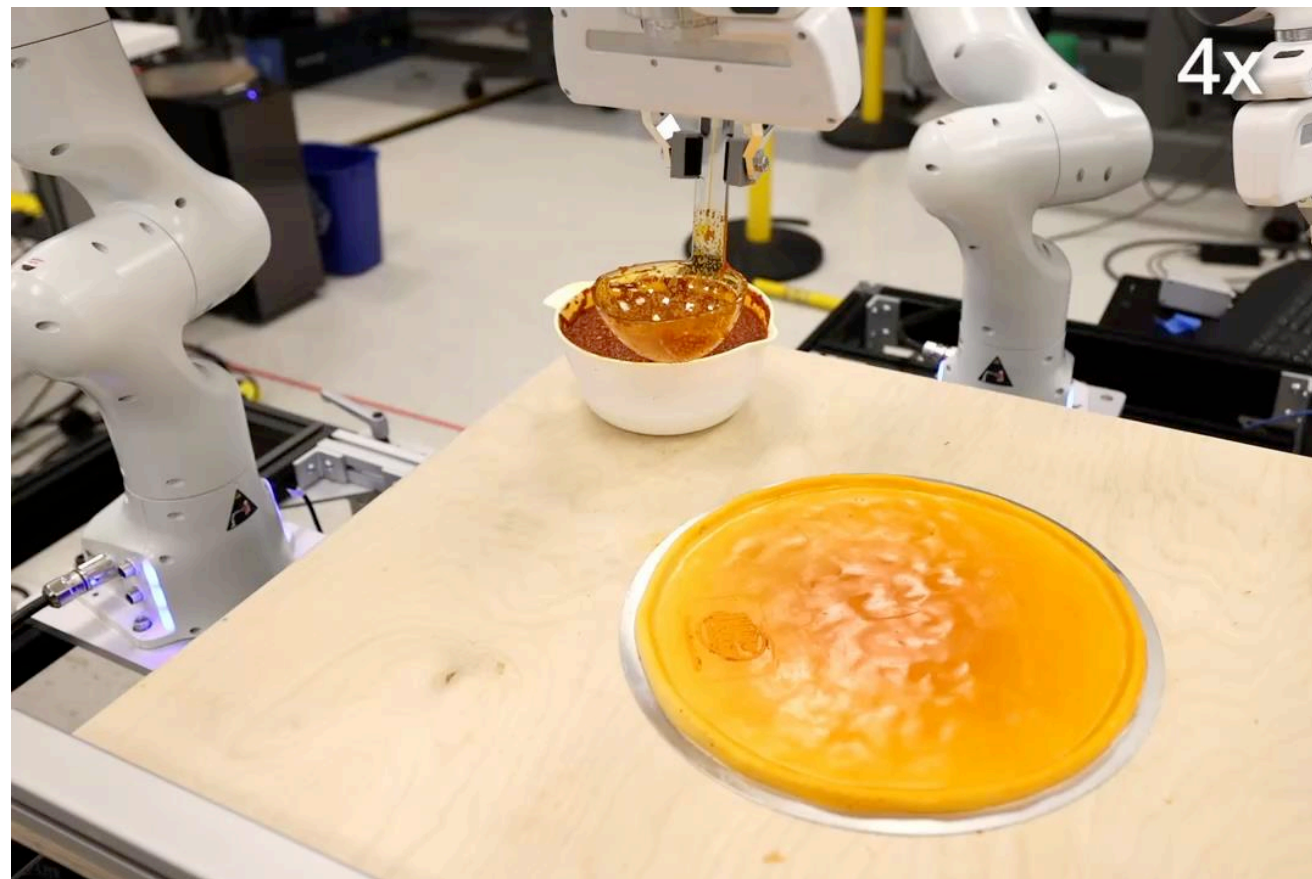
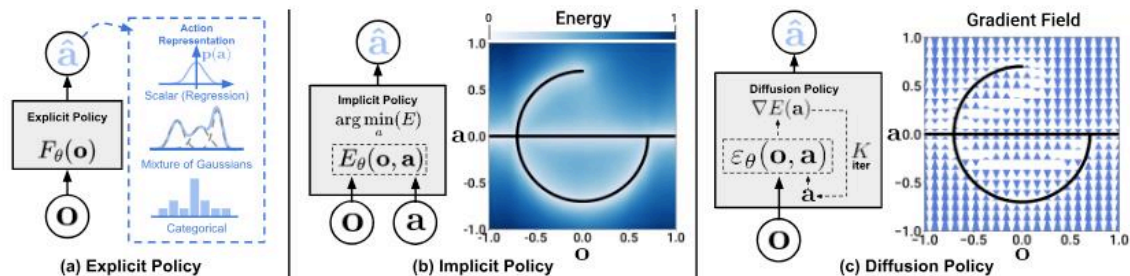


Where are We Today: Learning from Demonstrations

- ❑ Diffusion policy (Cheng et al.)

Diffusion Policy

Visuomotor Policy Learning via Action Diffusion

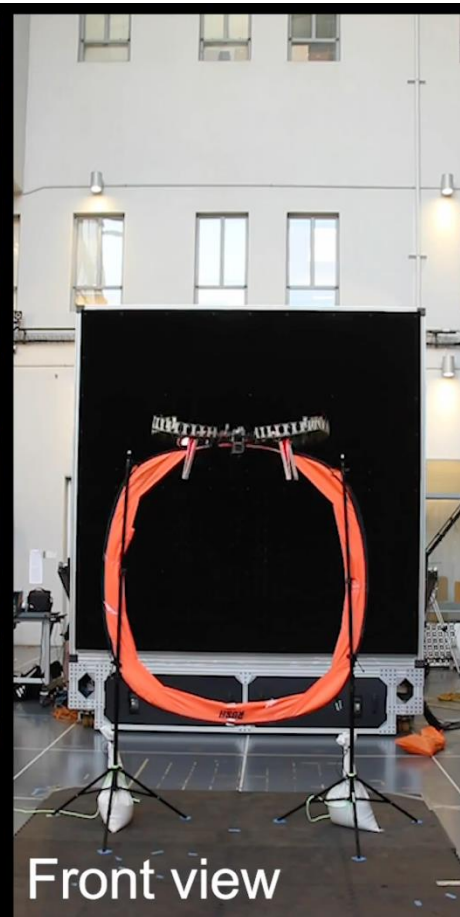


Where are We Today: Model-based RL

- ❑ No simulator. Collect data from real -> learn a model -> obtain a policy -> deploy
- ❑ Meta-learned dynamics model + online adaptive control



Our method can also generalize to the challenging time-varying wind condition.



Neural-Fly Enables Rapid Learning for Agile Flight in Strong Winds

Michael O'Connell[†], Guanya Shi[†], Xichen Shi,
Kamyar Azizzadenesheli, Anima Anandkumar, Yisong Yue, Soon-Jo Chung*

Division of Engineering and Applied Science, California Institute of Technology

[†] The first two authors contributed equally to this article. Alphabetical order.

* Corresponding authors. Email: sjchung@caltech.edu

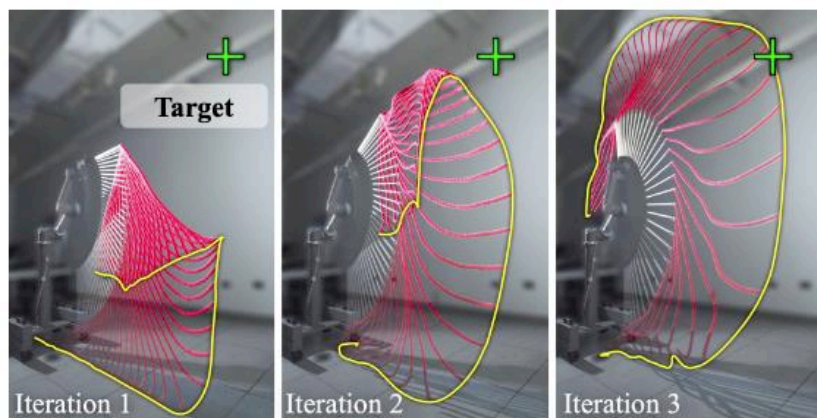
This is the accepted version of *Science Robotics* Vol. 7, Issue 66, eabm6597 (2022)

DOI: 10.1126/scirobotics.abm6597 Video: <https://youtu.be/TuF9teCZX0U>

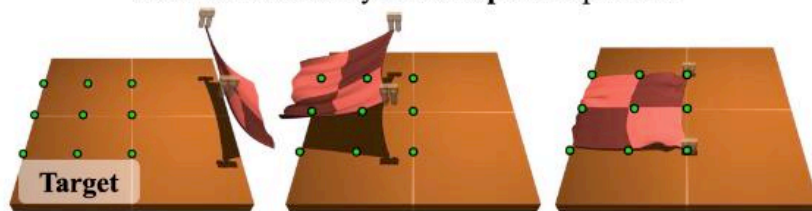
Data and training code: <https://github.com/aerorobotics/neural-fly>

Where are We Today: Model-based RL

- ❑ No simulator. Collect data from real -> learn a model -> obtain a policy -> deploy
- ❑ Whipping the rope to a target point via a learned “delta” dynamics model



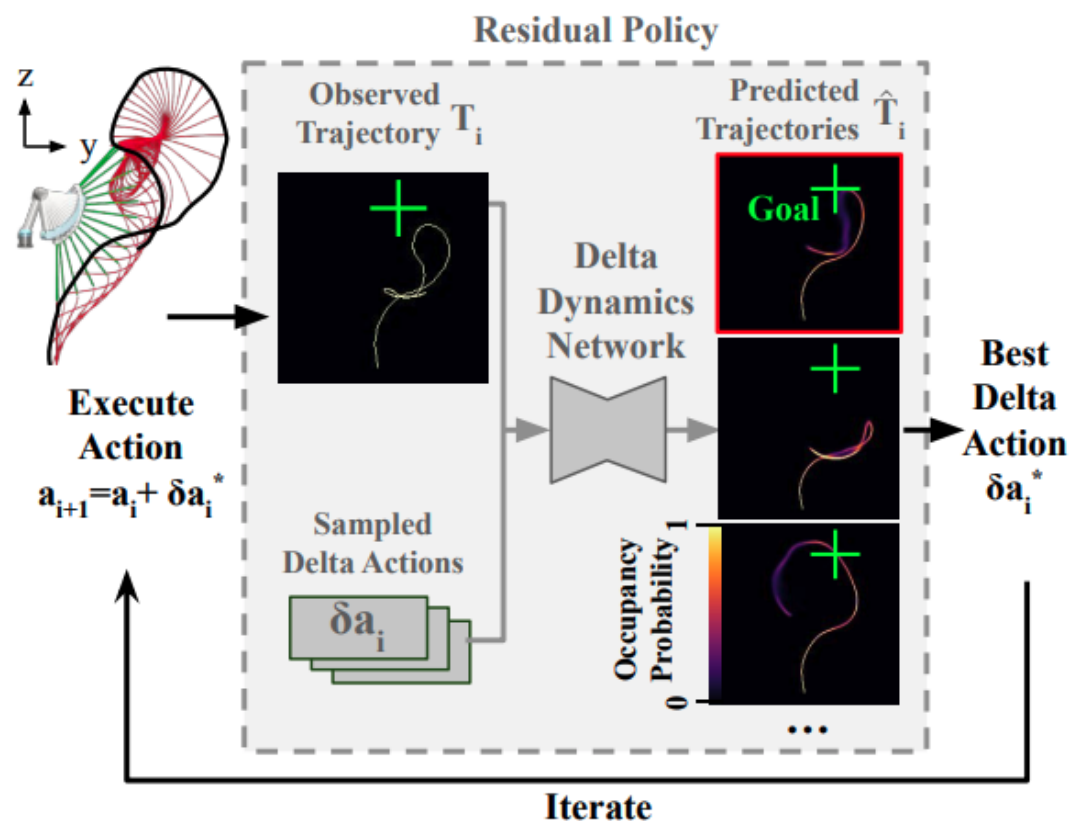
Goal-conditioned Dynamic **Rope** Manipulation



Goal-conditioned Dynamic **Cloth** Manipulation

Iterative Residual Policy

for Goal-Conditioned Dynamic Manipulation of Deformable Objects



Where are We Today: Model-based RL

- ❑ No simulator. Collect data from real -> learn a model -> obtain a policy -> deploy
- ❑ DNN dynamics + sampling-based MPC (MPPI)

Information Theoretic MPC for Model-Based Reinforcement Learning

Grady Williams, Nolan Wagener, Brian Goldfain, Paul Drews,
James M. Rehg, Byron Boots, and Evangelos A. Theodorou

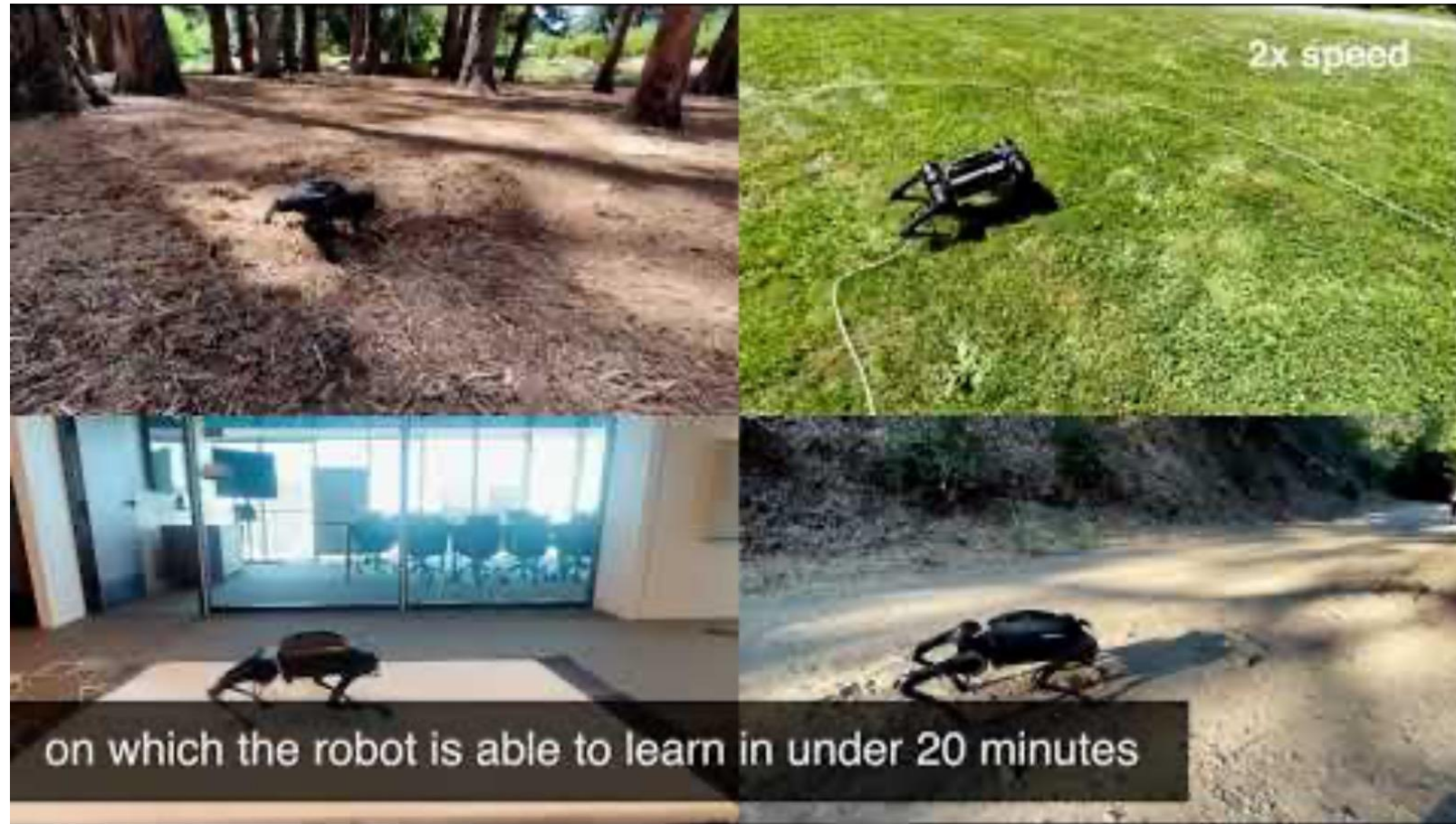


Where are We Today: Online Model-free RL

- ❑ No simulator. MFRL from scratch

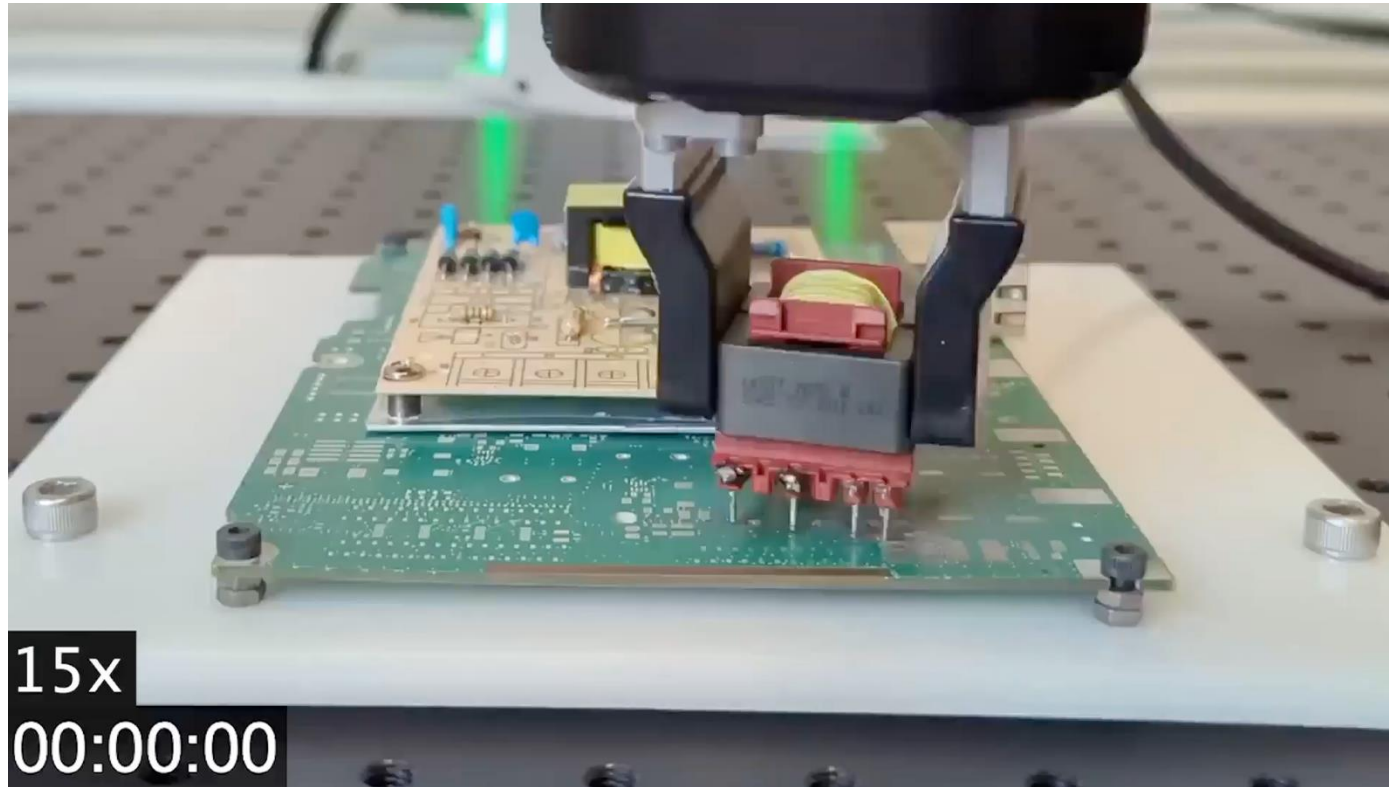
A Walk in the Park: Learning to Walk in 20 Minutes With Model-Free Reinforcement Learning

Laura Smith^{*1}, Ilya Kostrikov^{*1}, Sergey Levine¹
^{*}Equal contribution ¹Berkeley AI Research, UC Berkeley



Where are We Today: Online Model-free RL

- ❑ No simulator. MFRL from scratch



SERL: A Software Suite for Sample-Efficient Robotic Reinforcement Learning

ICRA 2024

[Featured on Hugging Face Daily Paper](#)

[Jianlan Luo](#)^{1*}, [Zheyuan Hu](#)^{1*}, [Charles Xu](#)¹, [You Liang Tan](#)¹, [Jacob Berg](#)², [Archit Sharma](#)³, [Stefan Schaal](#)⁴, [Chelsea Finn](#)³, [Abhishek Gupta](#)², [Sergey Levine](#)¹

^{*}Equal Contribution; ¹Department of EECS, University of California, Berkeley; ²Department of Computer Science, University of Washington;

³Department of Computer Science, Stanford University; ⁴Intrinsic Innovation LLC

Where are We Today: Offline RL

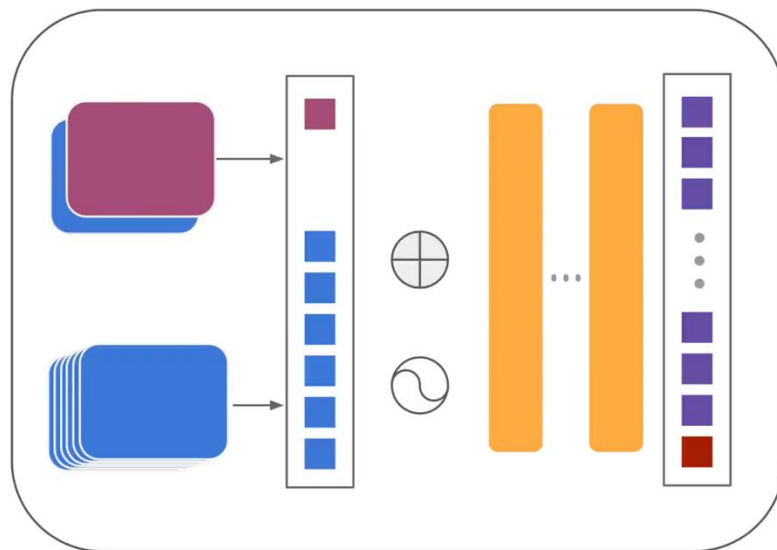
- ❑ No simulator. Learning from offline data.
- ❑ Difference from imitation learning: doesn't rely on expert data

General Navigation Models

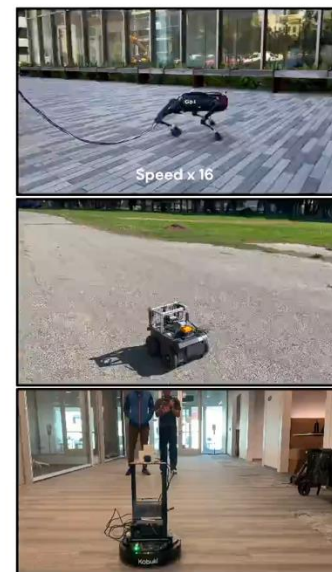
Dhruv Shah, Ajay Sridhar, Nitish Dashora, Catherine Glossop,
Kyle Stachowicz, Arjun Bhorkar, Kevin Black, Noriaki Hirose, Sergey Levine



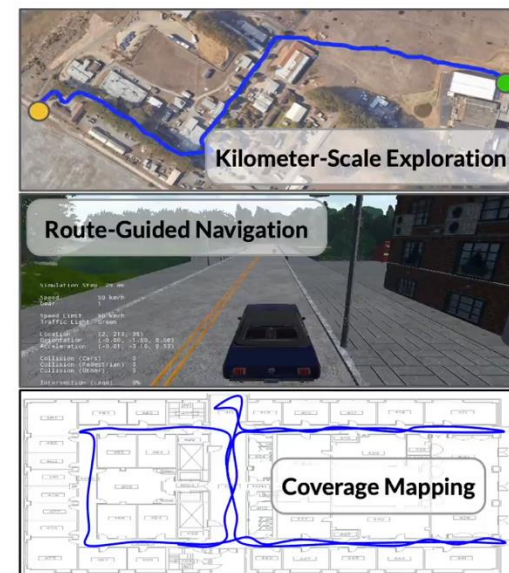
Diverse Training Data



General Navigation Models



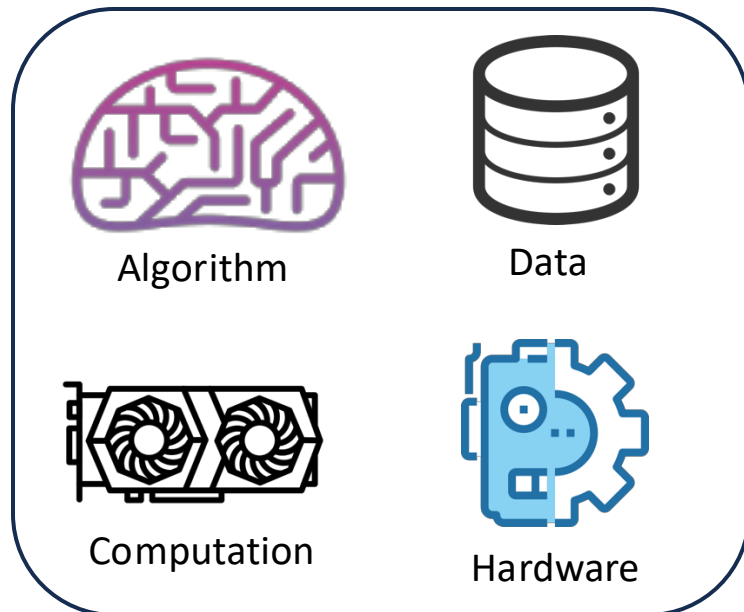
Deploy Zero-Shot



Adapt to Downstream Tasks

We Have Mentioned these Robot Learning Methods:

- ❑ In this class, we will learn how these methods work and more!
 - Model-free reinforcement learning (RL)
 - Sim2Real and Real2Sim2Real
 - Online adaptation
 - Imitation learning with demonstrations
 - Model-based RL
 - Offline RL



How far are we?



Embodied AGI in the real physical world



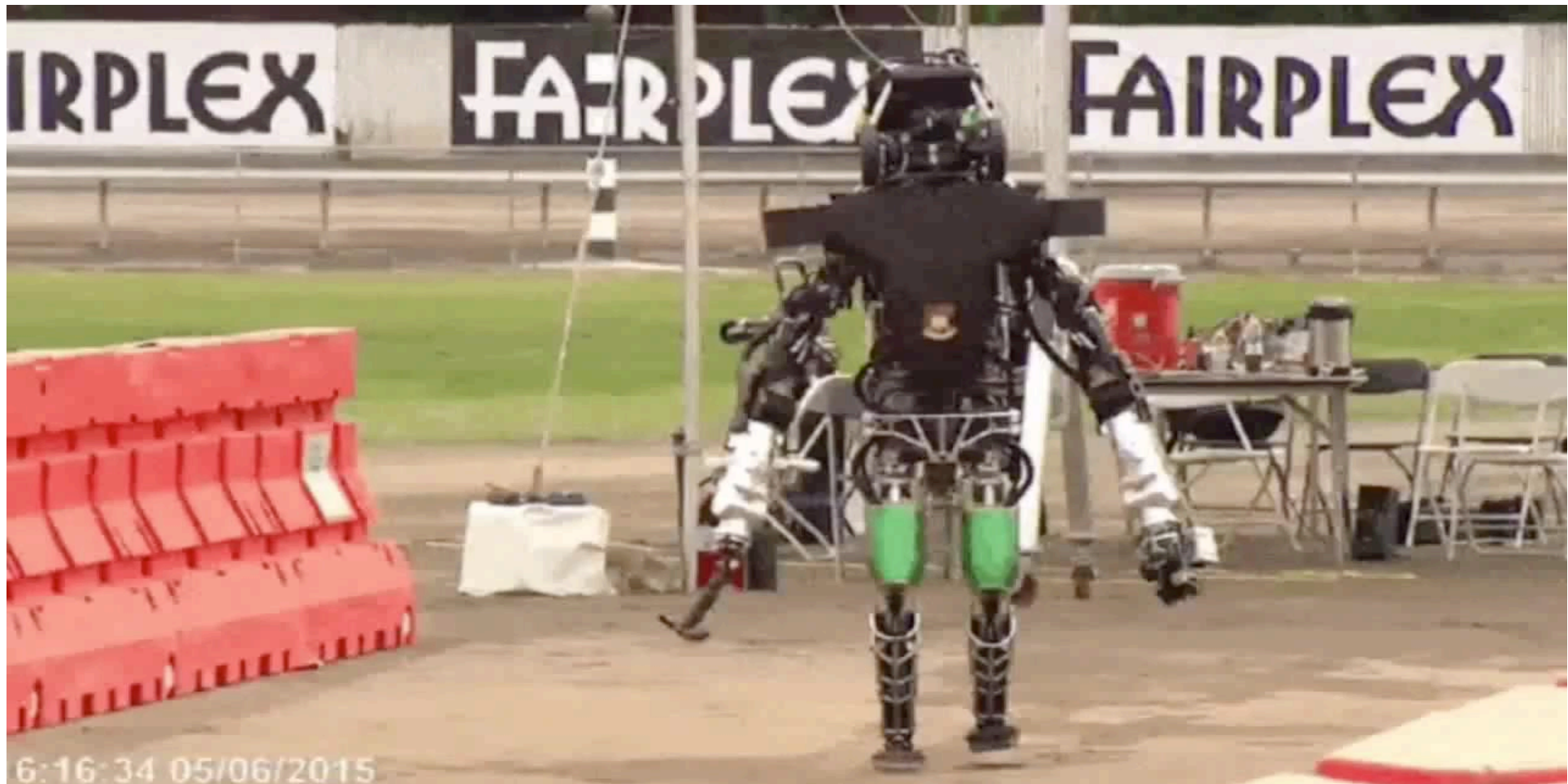
Still Many Failures...

- ❑ These tasks seem “very simple” for humans



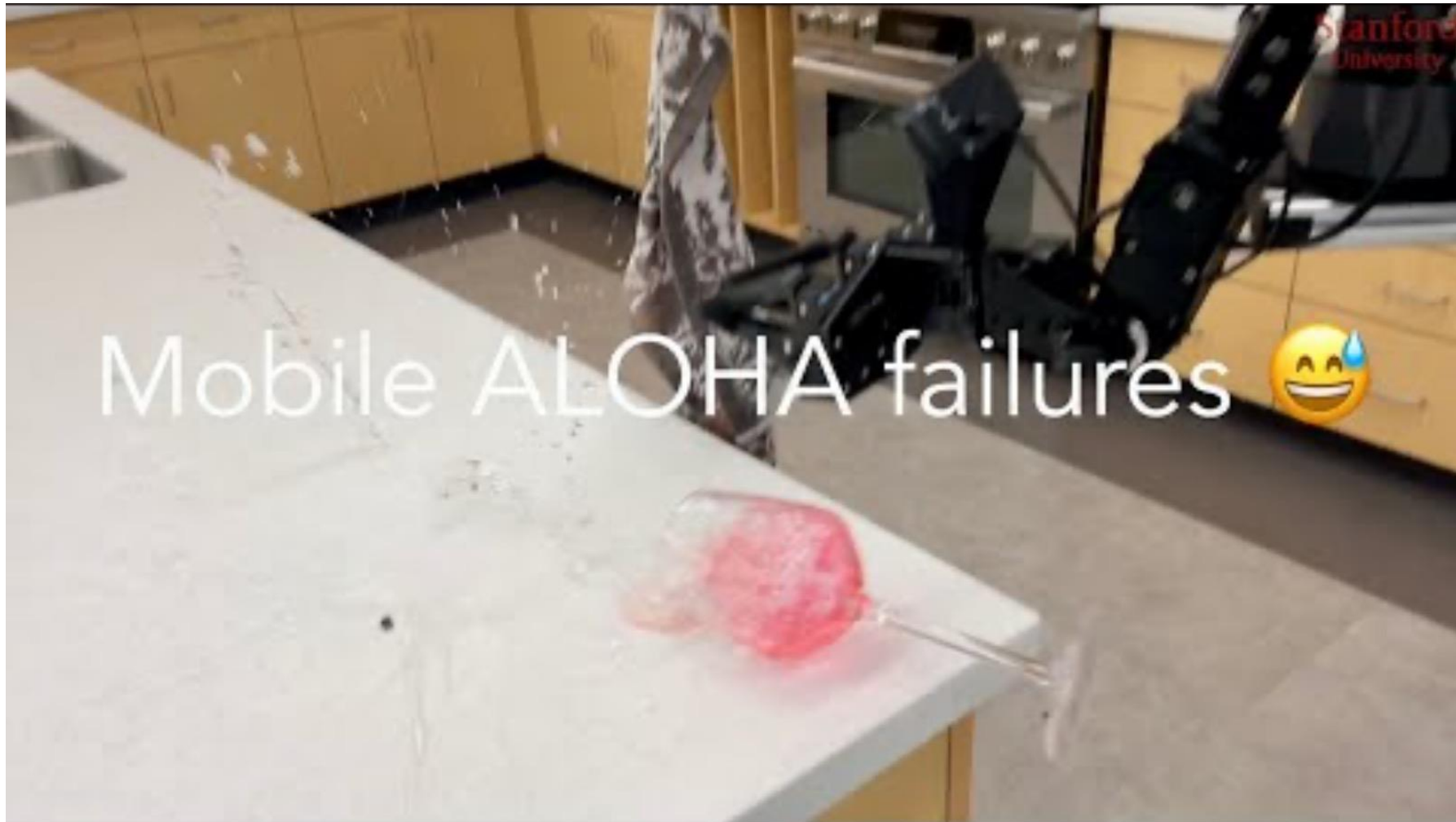
Still Many Failures...

- ❑ These tasks seem “very simple” for humans
- ❑ DARPA challenge 2015:



Still Many Failures...

- ❑ These tasks seem “very simple” for humans
- ❑ Failures in Mobile ALOHA:



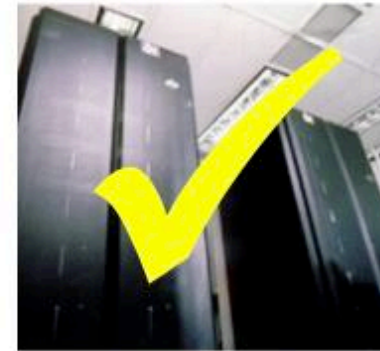
Moravec's Paradox

"The hard problems are easy and the easy problems are hard."

What to move where



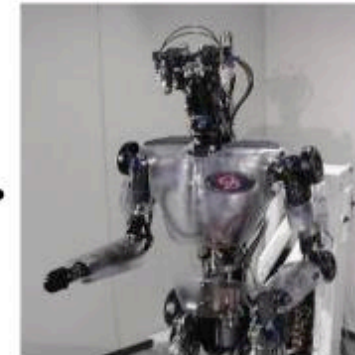
VS.



Moving



VS.



Common Belief



Starting of Life

Apes

Humans

Language



3.5B Years

~18M Years

~50K-500K Years

Moravec's Paradox



Hans P. Moravec
Principal Research Scientist
Robotics Institute
Carnegie Mellon University

“It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to *perception* and *mobility*.” (1988)

Moravec's Paradox: Why?

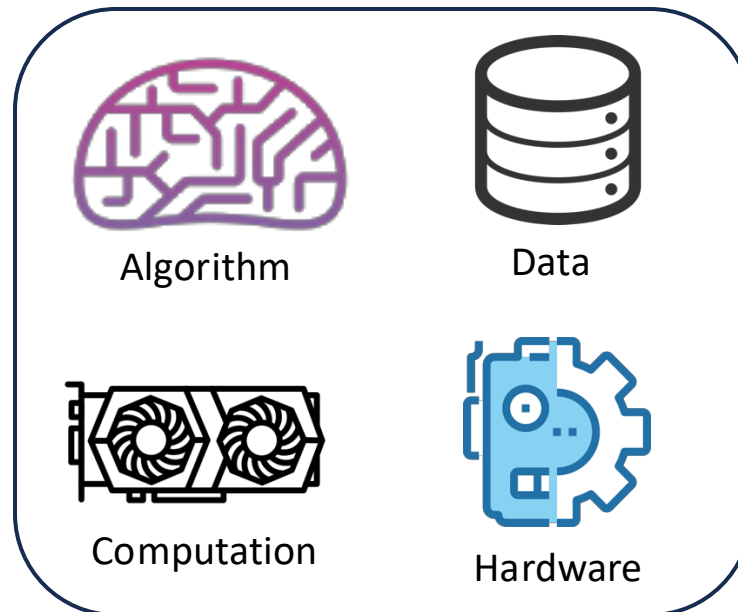
- ☐ How many of you think the difficulty of motor control is from the hardware?
 - I.e., humans have much better “motors” than robots
- ☐ The answer is likely no! When robots are with human brains(human teleoperations):



OmniH2O: Universal and Dexterous Human-to-Humanoid
Whole-Body Teleoperation and Learning
(<https://omni.human2humanoid.com/>)

Moravec's Paradox: Why?

- ❑ The ALOHA system is relatively cheap (<32K) and not too many DoFs (degree of freedom)
 - 2 (base) + 12 (two arms) + 2 (two grippers)
- ❑ Can we conclude that algorithm/data/computation is more important than hardware?



Moravec's Paradox: Why?

- ❑ Not that simple! The coupling between different components is very important
- ❑ The human teleoperator knows and leverages the hardware structure
 - Perception-planning-control-actuation in a closed loop!



Moravec's Paradox: Why?

- ❑ Another example: drone racing
 - It took the team >7 years to beat the human champion
 - Human racers: huge delay ($>0.2s$), “offboard” control, ...
 - AI barely wins in a specific known track (it can not generalize)
- ❑ Human racers: perception-planning-control-actuation in a closed loop!



Alex Vanover

2019 DRL World Champion



Thomas Bitmatta

2019 Multi GP Champion



Marvin Schäpper

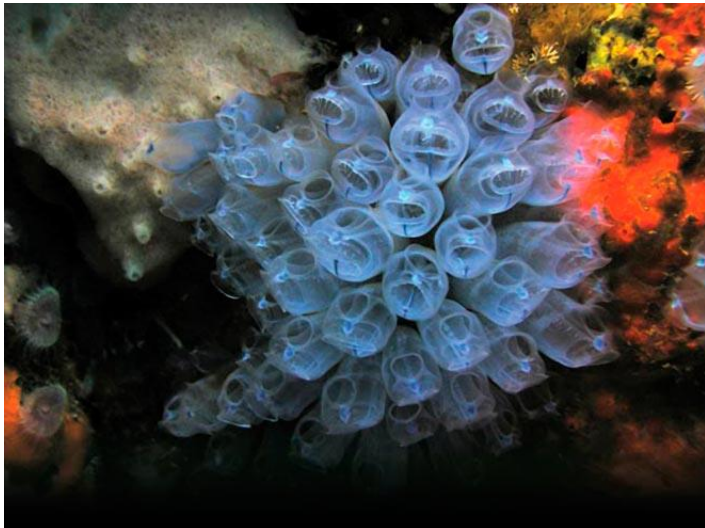
3X Swiss Champion



Some Cognitive Science Perspectives

- ❑ Perception and action are coupled

“We have brain for one reason and one reason only, and that’s to produce adaptable and complex movements.” -- Daniel Wolpert



Sea squirts swim through ocean in early life. But when they decide not to move after implanting on a rock, they digest their own brain and nervous system.

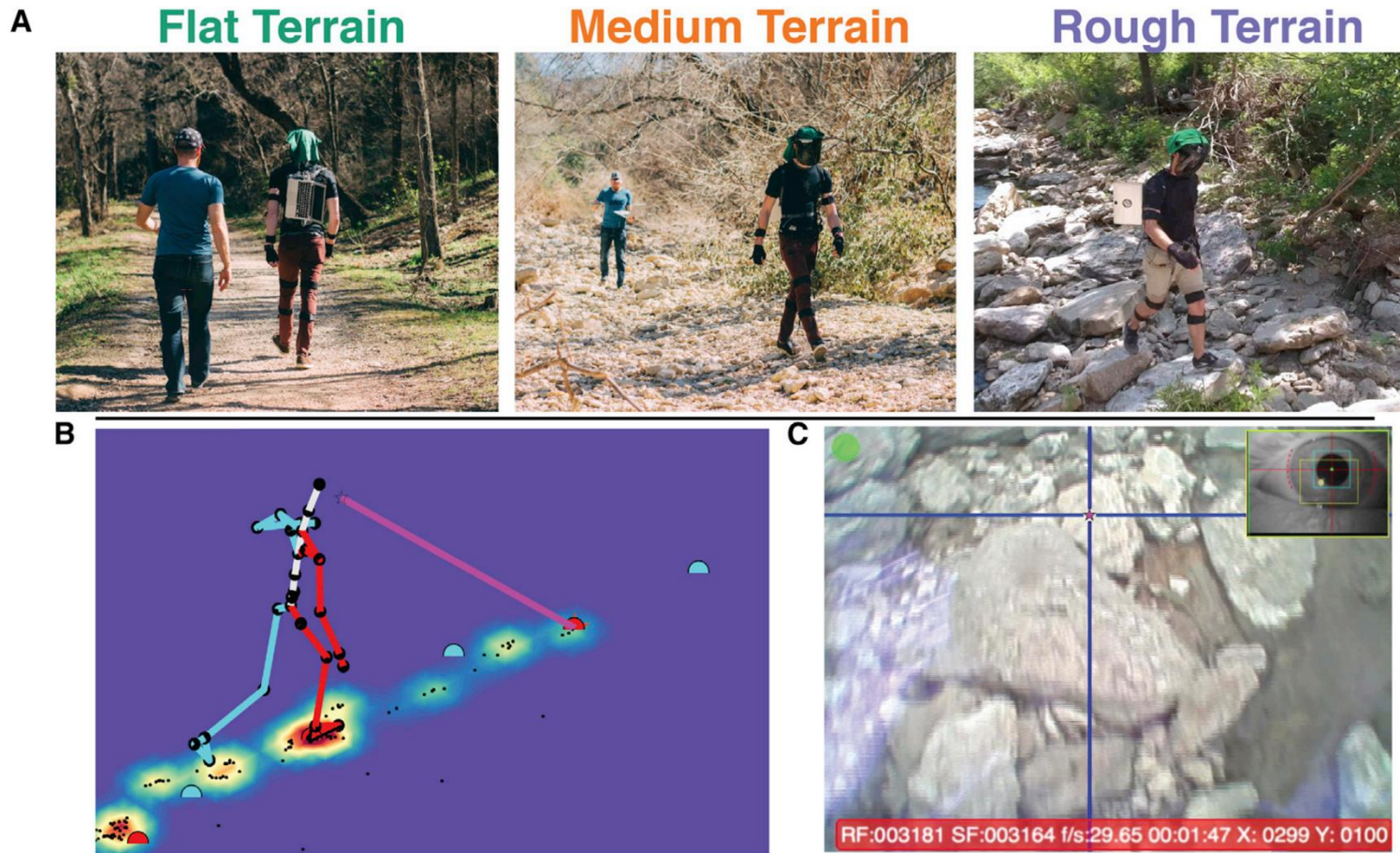
Some Cognitive Science Perspectives

- ❑ Perception and action are coupled
- ❑ Example: control-perception loop of a crow



Some Cognitive Science Perspectives

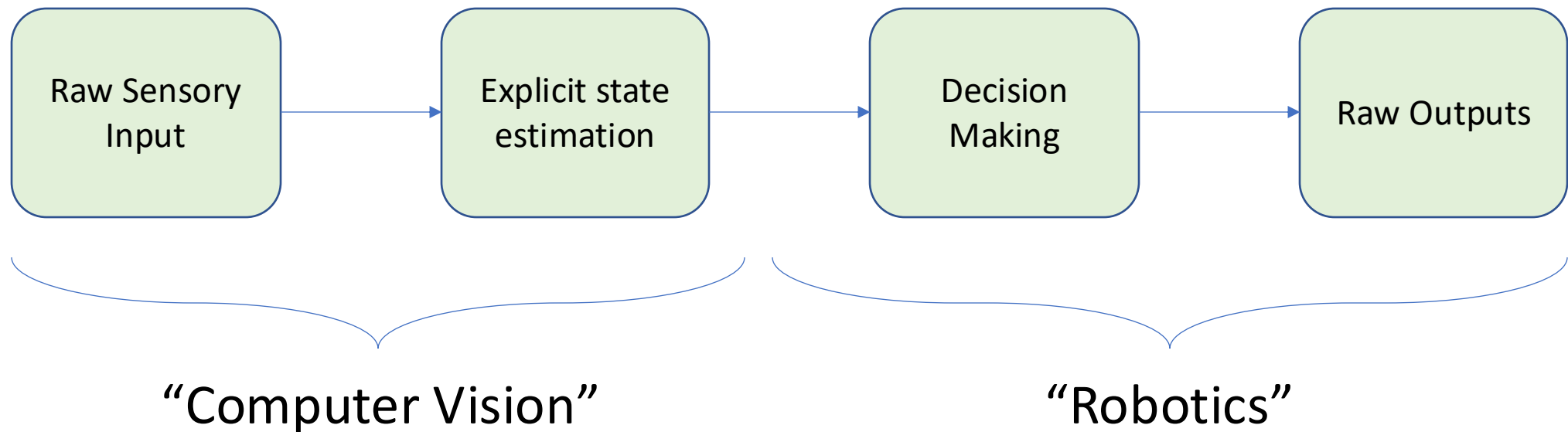
- ❑ Perception and action are coupled
- ❑ Example: gaze and the foot placement



Matthis, Jonathan Samir, Jacob L. Yates, and Mary M. Hayhoe. "Gaze and the control of foot placement when walking in natural terrain." *Current Biology* (2018)

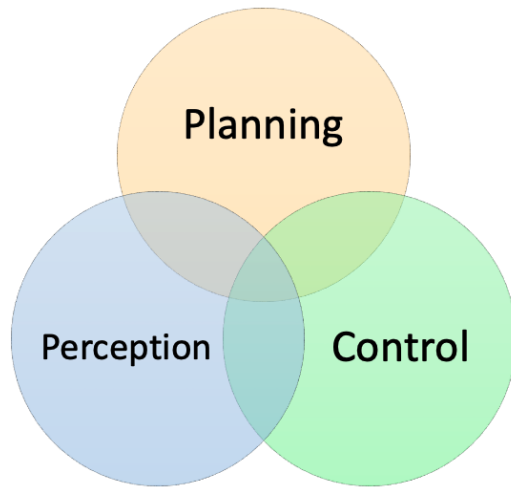
However, No Easy Way to Unify!

- ❑ Perception: high-dimensional **input**
 - How to get to the low-dimensional manifold of meaningful things from raw input
- ❑ Control: high-dimensional **output**
 - Exponential search space that grows with number of steps and number of actions
- ❑ Perception + control: high-dimensional input AND output

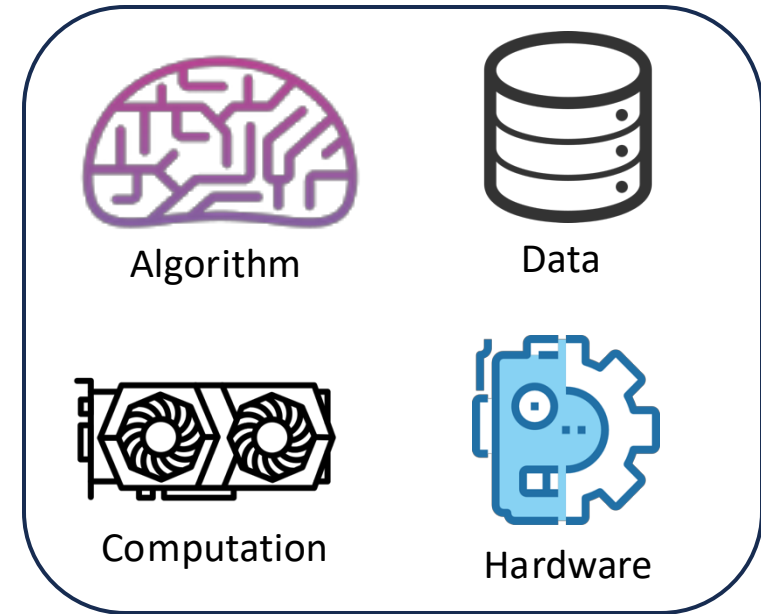


This is Where Robot Learning Helps!

- ❑ (1) Robot learning can unify (or at least, let these modules talk with each other):



by



- ❑ (2) Robot learning can also improve each module especially when it is embodied!
- ❑ We have made much more progress in (2) than (1)

Today's Agenda

- ☐ What is robot learning and what is its uniqueness?
- ☐ What is the goal of robot learning?
- ☐ Where are we today? How far are we from the goal?
- ☒ **Course structure and course objective**
- ☐ Relation to other CMU courses
- ☐ Applications of sequential decision making in other domains

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☒ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Supervised Learning



Input x

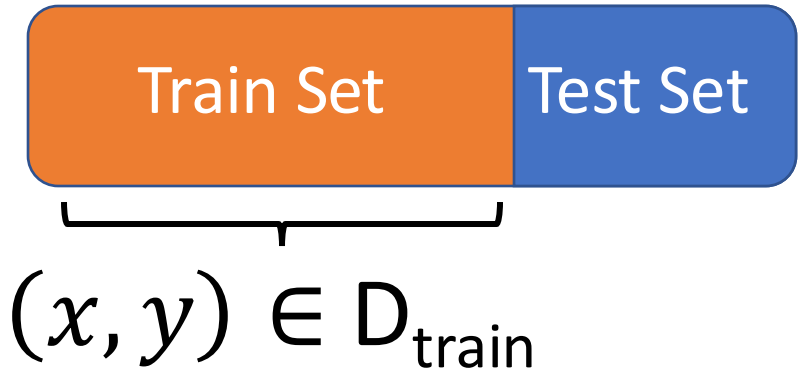


Model $f_{\theta}()$



“Labrador”

Output
 $y = f_{\theta}(x)$



Supervised Learning



Input x

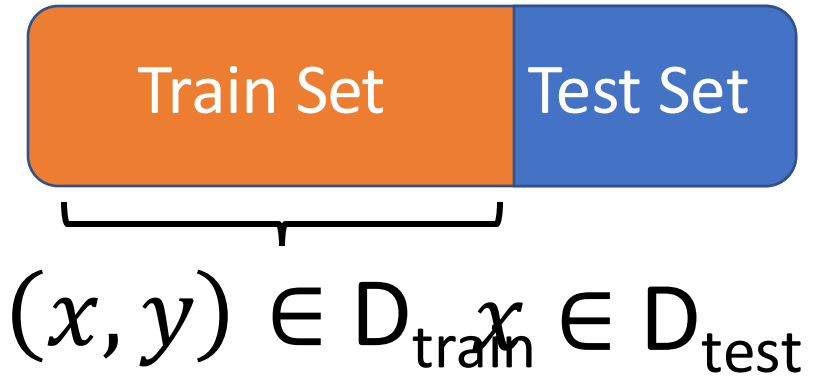


Model $f_{\theta}()$



“Labrador”

Output
 $y = f_{\theta}(x)$



Machine/Deep Learning Refresher

- ❑ Robot learning, by definition, is highly interdisciplinary!
- ❑ Robot learning can use all possible machine/deep learning techniques

- ❑ The goal of refresher: briefly recap machine/deep learning concepts, especially those useful for robotics
 - Paradigm: supervised regression/classification, unsupervised, generative model, non-parametric learning, etc
 - Architecture: MLP, ResNet, CNN, GNN, RNN/LSTM, transformer, etc
 - Optimization: GD, SGD, etc
 - Other topics: uncertainty quantification, verification, etc

- ❑ Disclaimer: It is NOT a machine learning course! We will only briefly (and intuitively) introduce these concepts. Check out the DL textbook (link on the website) for more!

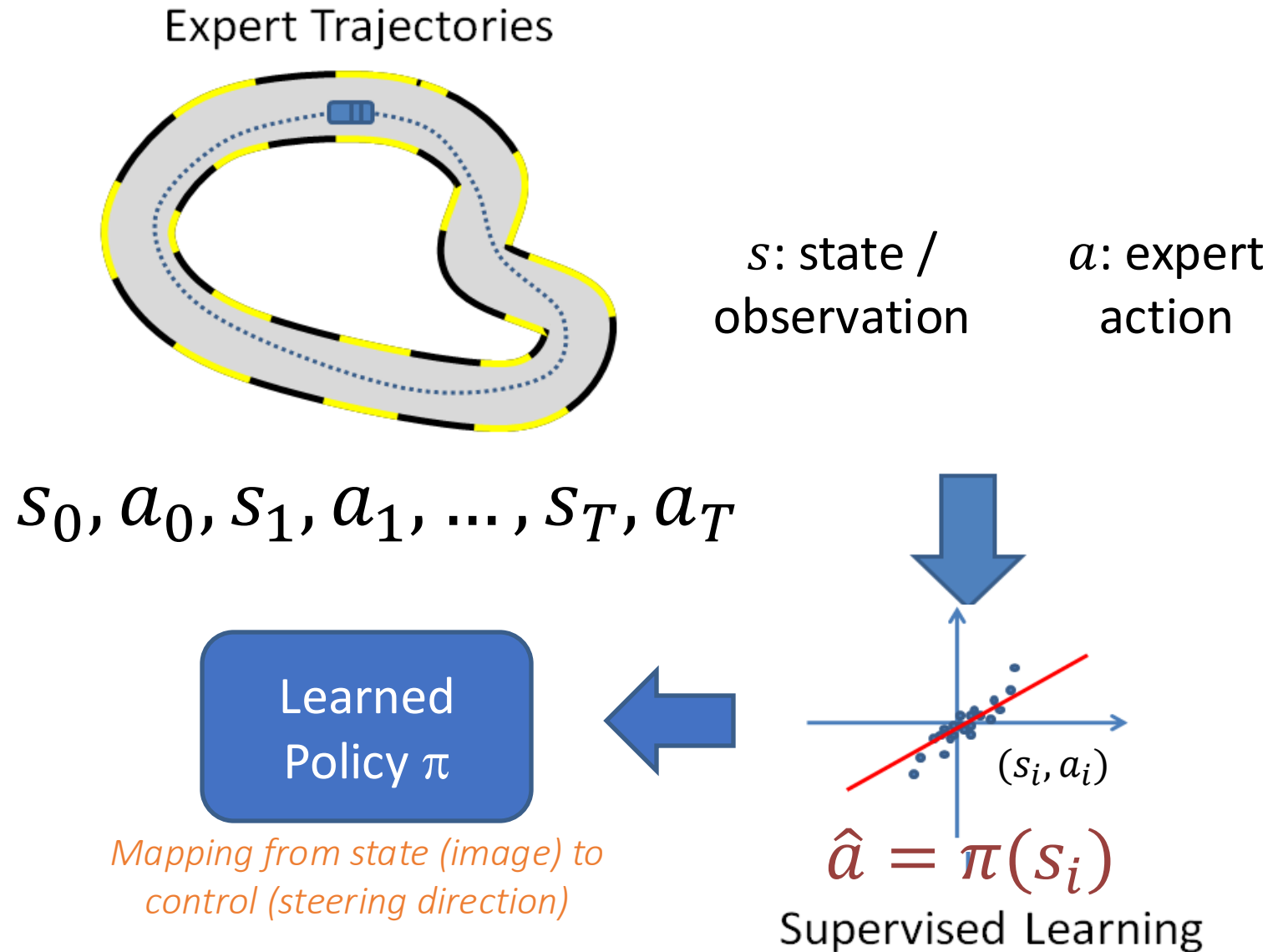
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☒ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

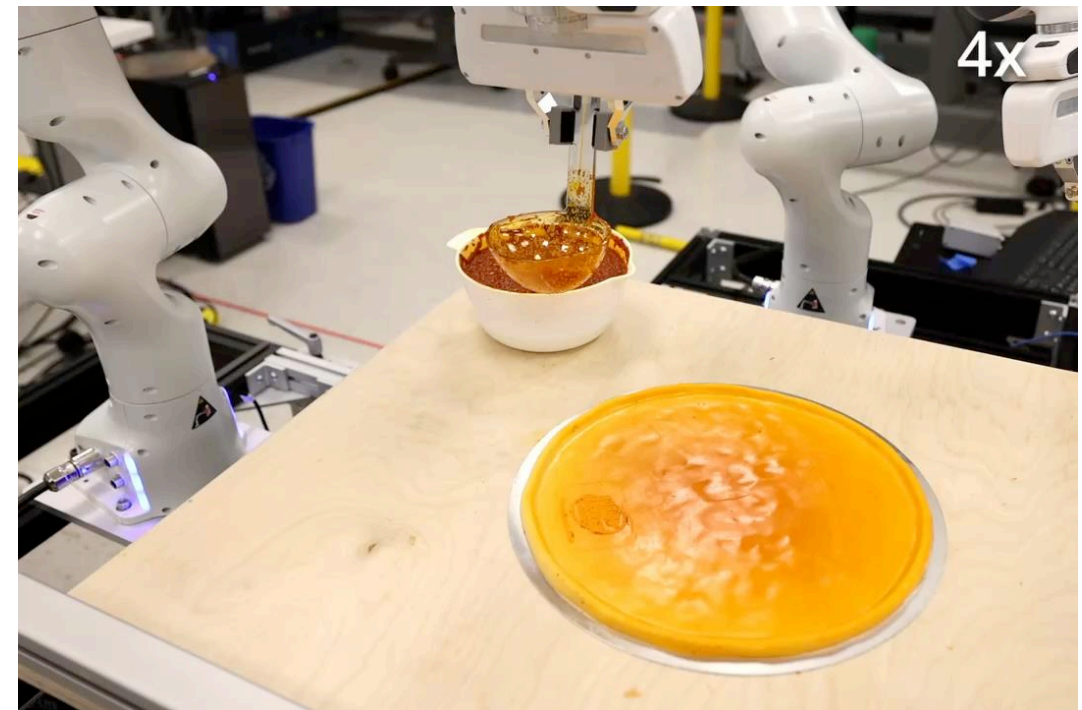
Imitation Learning

- ❑ How is learning from demonstrations different from standard supervised learning?
- ❑ How to design efficient and effective algorithms?



Imitation Learning

- ❑ You will learn how these demos work
 - Left: Mobile ALOHA; Right: Diffusion policy.



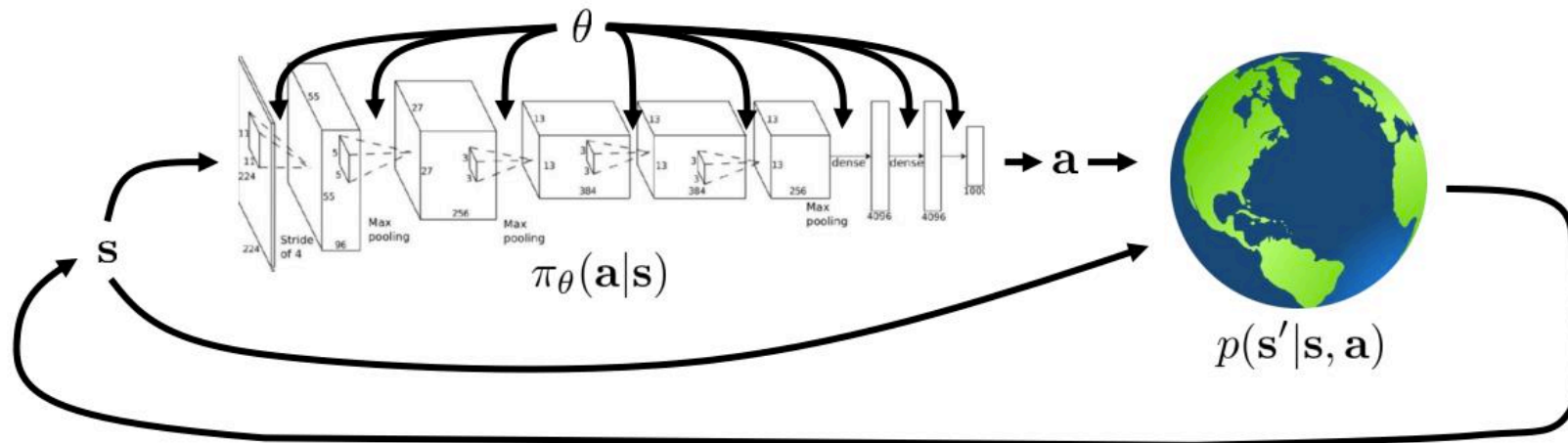
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☒ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Model-free Reinforcement Learning

- ❑ Learning from directly interacting with the environment
- ❑ Learning from trial-and-errors



$$\underbrace{p_\theta(s_1, a_1, \dots, s_T, a_T)}_{\pi_\theta(\tau)} = p(s_1) \prod_{t=1}^T \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t)$$

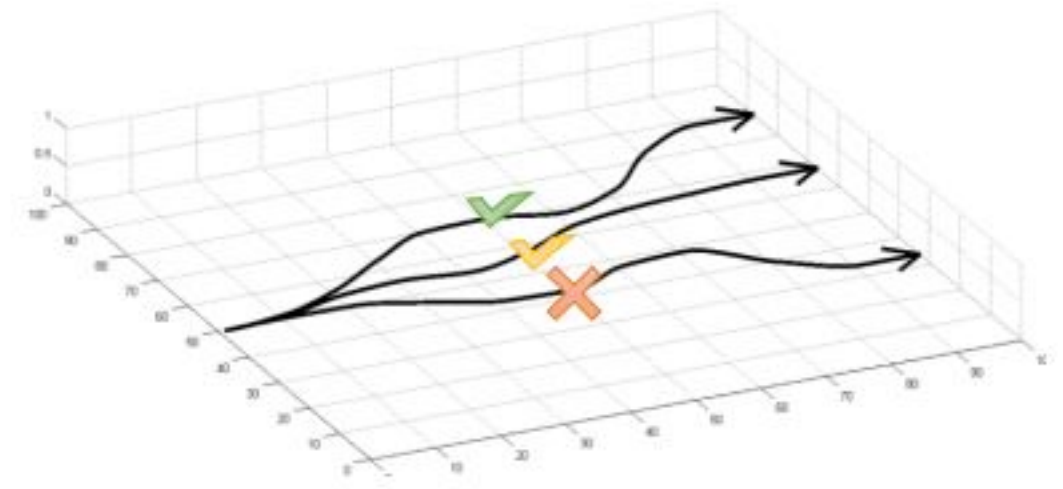
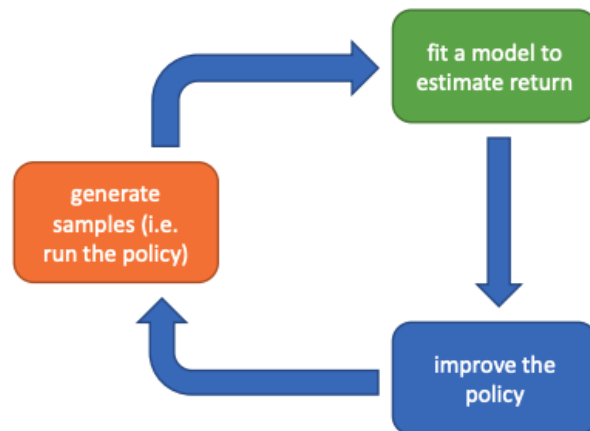
$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(s_t, a_t) \right]$$

Model-free Reinforcement Learning

- ❑ Example: REINFORCE (the basic policy gradient algorithm)
 - Directly interact with the environment to improve the policy
 - Trail and error: make “good stuff” more likely and “bad stuff” less likely
 - No model in the algorithm

REINFORCE algorithm:

1. sample $\{\tau^i\}$ from $\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ (run the policy)
2. $\nabla_\theta J(\theta) \approx \sum_i (\sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^i|\mathbf{s}_t^i)) (\sum_t r(\mathbf{s}_t^i, \mathbf{a}_t^i))$
3. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$



Model-free Reinforcement Learning

- ❑ We will learn:
 - RL basics (policy iteration, value iteration)
 - Valued-based methods: Q-learning and variants
 - Policy-based methods: Policy gradient methods
 - Actor-critic methods
 - Practical and advanced (deep) RL algorithms

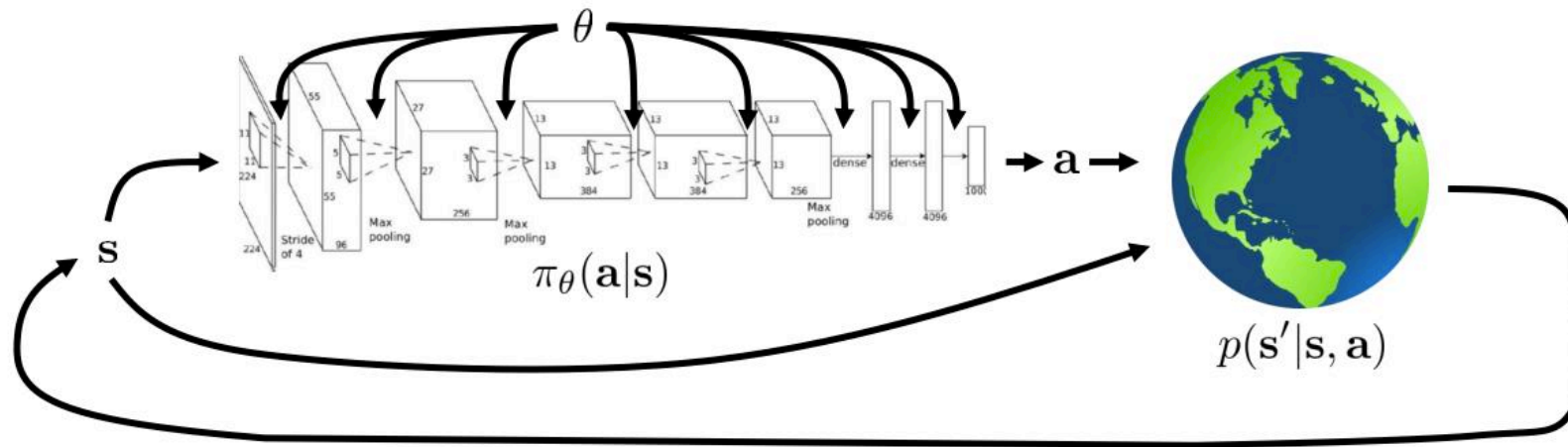
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☒ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Model-based Control and RL

- Build/learn a model (transition dynamics, “world model”) and then control



$$\underbrace{p_\theta(s_1, a_1, \dots, s_T, a_T)}_{\pi_\theta(\tau)} = p(s_1) \prod_{t=1}^T \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t)$$

$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(s_t, a_t) \right]$$

model-free RL assumes it is unknown
(don't even attempt to learn it)

Model-based Control and RL

□ We will learn:

- Model-based control (with known model): PID, Lyapunov, etc.
- Optimal control (with known model): LQR, iLQR, DDP, SQP, etc.
- Model-based reinforcement learning (need to learn dynamics): CEM, MPPI, Neural-Control, etc.
- Deep MBRL: Dreamer, TD-MPC, etc.

$$\begin{array}{ll} \min_{x,u} & \sum_{t=1}^T c_t(x_t, u_t) \\ \text{s. t.} & \left\{ \begin{array}{l} x_{t+1} = f_t(x_t, u_t) \\ x_t \in X_t \\ u_t \in U_t \end{array} \right. \end{array}$$

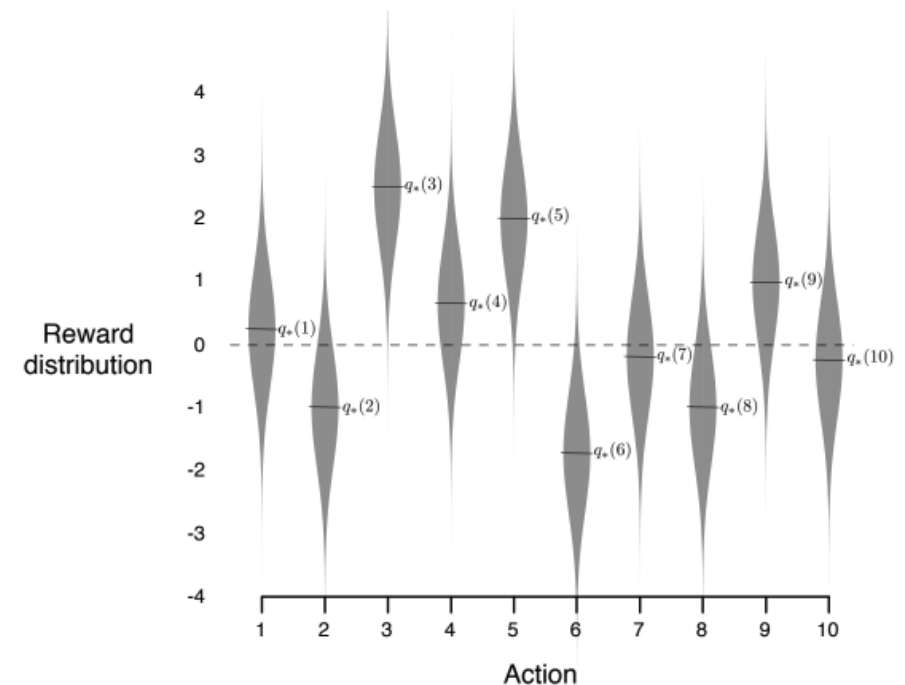
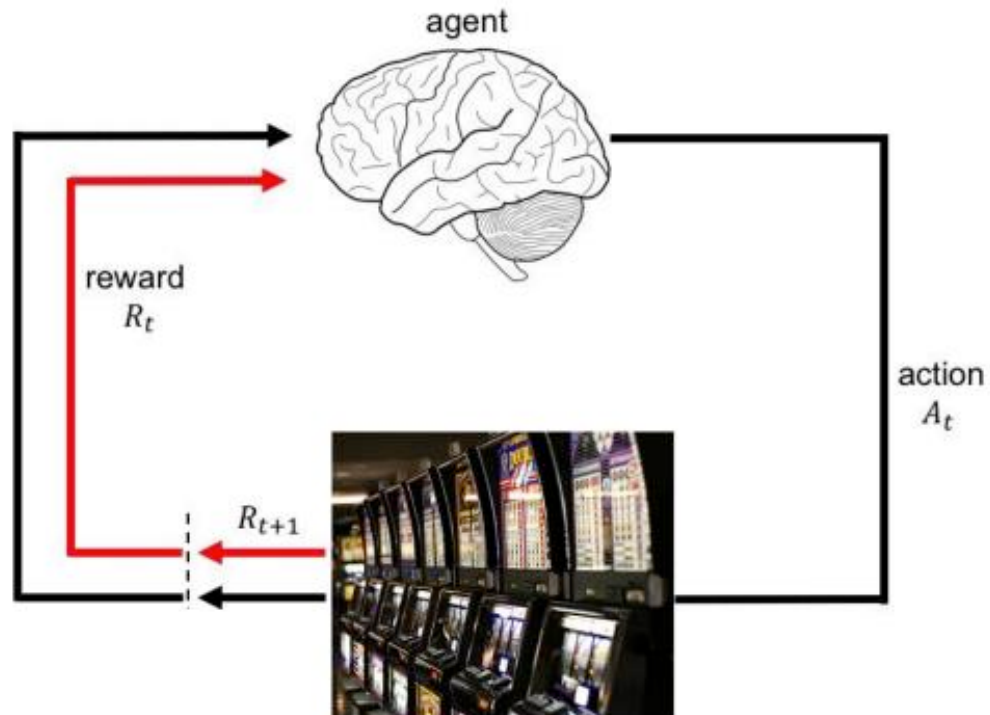
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☒ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Bandit and Exploration

- ❑ Bandit is the simplest RL problem (no state)
 - Therefore, it allows us to more fundamentally reason exploration v.s. exploitation
 - Many algorithmic principles from bandits can be transferred to RL
- ❑ We will also introduce preference-based learning (no reward, only relative preference)



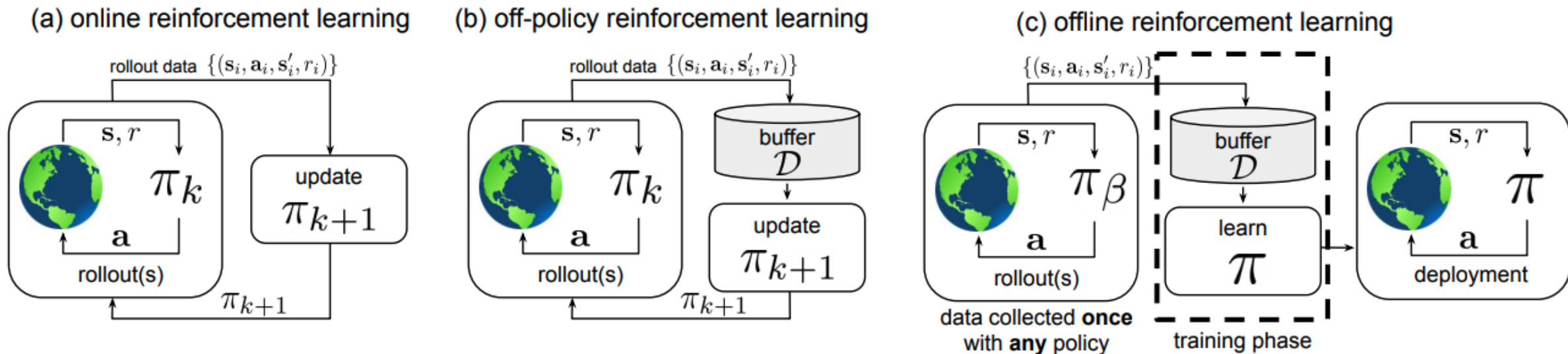
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☒ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Reinforcement Learning from Offline Data

- ❑ Motivation: Online RL relies exclusively on interaction
 - Expensive and unsafe
- ❑ RL from offline dataset? Question:
 - How to learn?
 - How is it different from imitation learning?
- ❑ We will also talk about inverse RL



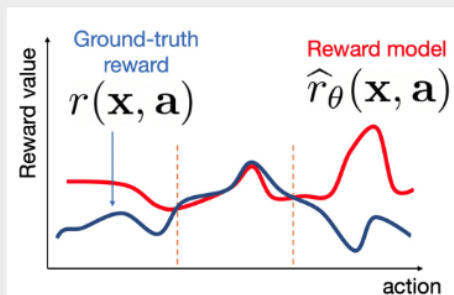
Guest Speaker 1: Aviral Kumar

In-person! Assistant Professor at CMU CSD/MLD (<https://aviralkumar2907.github.io/>).

❑ Topic: Offline RL: algorithms, extensions, and applications

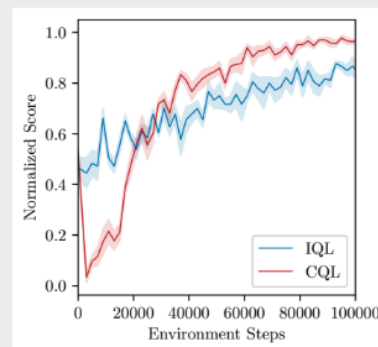
Today's Lecture: Several Facets of Offline RL

Algorithms for Offline RL



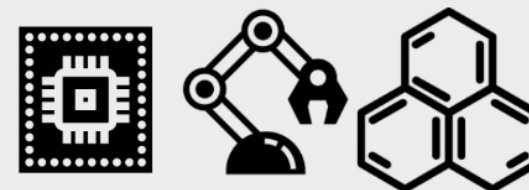
Conservative Q-learning,
Model-based offline RL,
large-scale results

Extensions of Offline RL



Online fine-tuning,
learning with unlabeled data

Real-World Applications



Chip Design & robotic pre-
training and fine-tuning

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

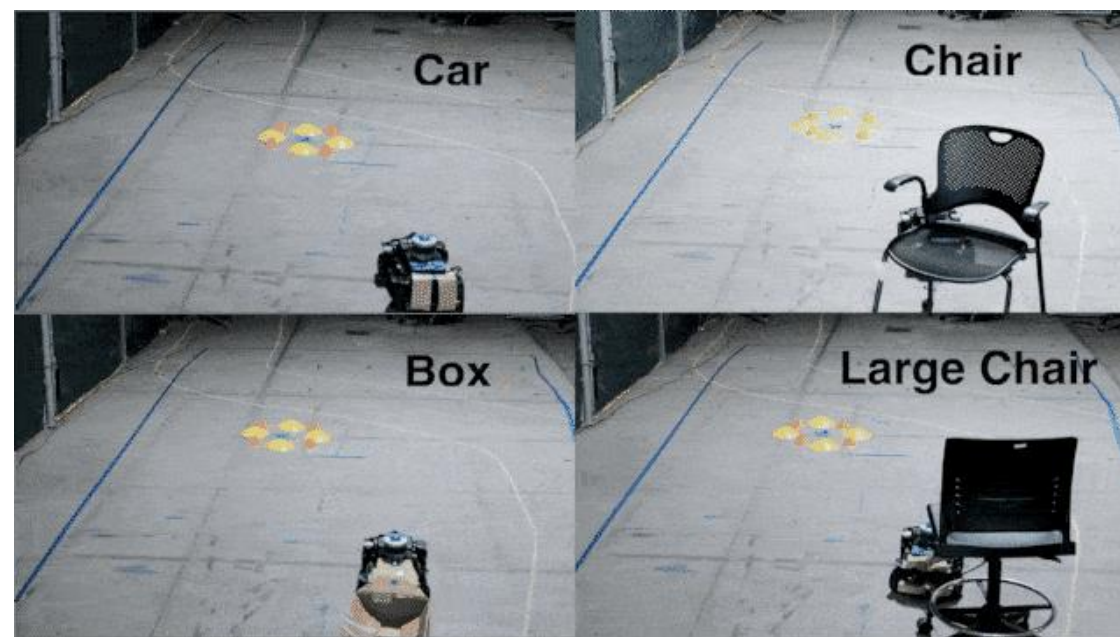
- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☒ Topic group 8: Specialized topics (**one guest lecture**)
- ☐ Project presentation

Specialized Topics

- ❑ Topic 1: Safe RL and safe robot learning
 - Collision-free, smooth, stable, etc.
- ❑ Topic 2: Multi-task/adaptive/transferrable robot learning
 - Adapt to different dynamics (e.g., different robots)
 - Adapt to different tasks (e.g., goal-conditioned RL)



Wabersich et al., Data-Driven Safety Filter, 2023



Xiao and He et al., Safe Deep Policy Adaptation

Specialized Topics

- ❑ Topic 3: Robot simulator and sim2real
 - Introduce various simulators
 - Introduce various sim2real (and real2sim2real) methods

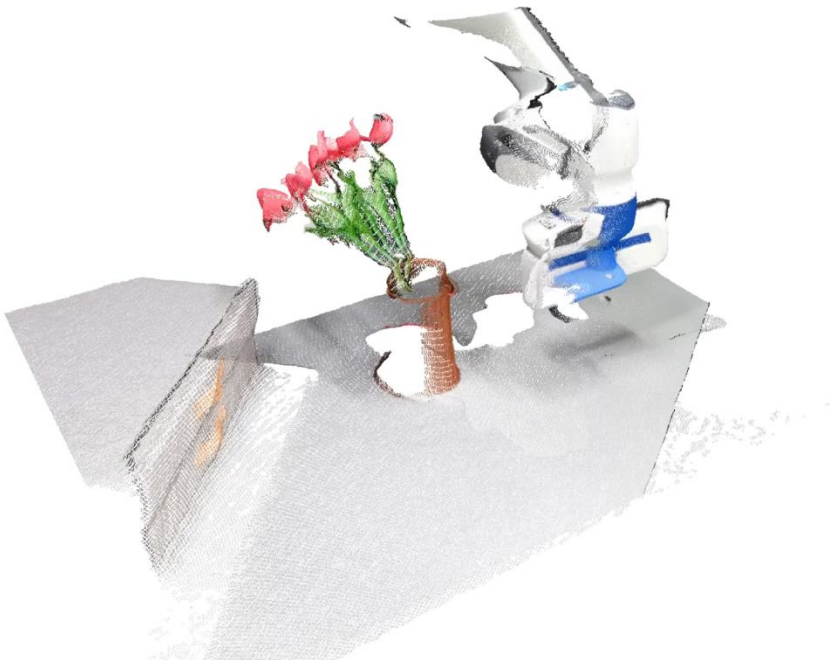


NVIDIA Isaac Gym



Specialized Topics

- ❑ Topic 4: Large Language Models (LLMs) and Vision Language Models (VLMs)
 - Briefly introduce LLMs and VLMs
 - Applications in robotics: from high-level task planning (e.g., VoxPoser in the left) to low-level control (Eureka in the right)



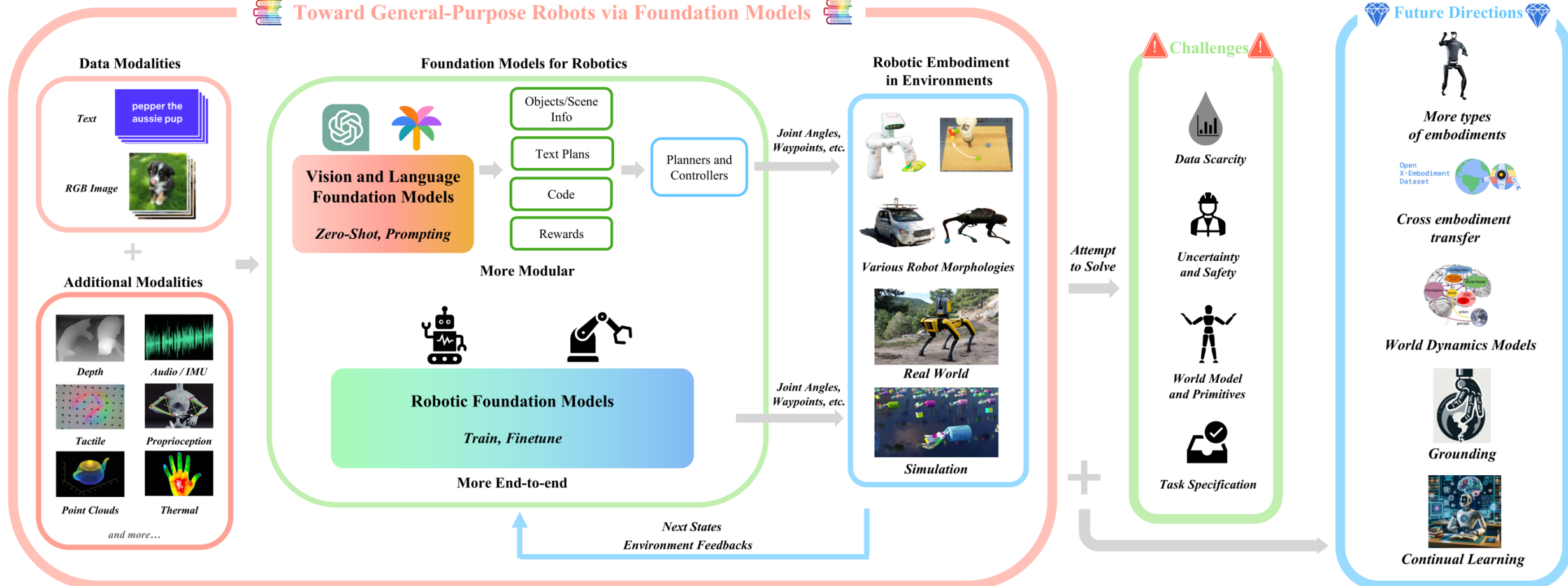
Guest Speaker 2: Yonatan Bisk

In-person! Assistant Professor at CMU LTI (<https://yonatanbisk.com/>).

❑ Toward General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis

❑ <https://robotics-fm-survey.github.io/>

Toward General-Purpose Robots via Foundation Models



Course Objective

- ☐ Study both fundamentals and advanced topics towards the goal of building *Embodied Intelligence*
- ☐ Learn to critically read and analyze research papers
- ☐ Brainstorm new ideas and follow-up directions
- ☐ Sell your ideas to other people: give a 5-min talk and make a poster
- ☐ Learn to convey complex ideas in a research discussion in your final project report

Today's Agenda

- ☐ What is robot learning and what is its uniqueness?
- ☐ What is the goal of robot learning?
- ☐ Where are we today? How far are we from the goal?
- ☐ Course structure and course objective
- ☒ **Relation to other CMU courses**
- ☐ Applications of sequential decision making in other domains

Relation to Other CMU Courses

- ❑ 10-703 Deep Reinforcement Learning (Katerina Fragkiadaki)
 - Focus on generic sequential decision making
 - 16-831 focuses on robotics
 - Recommend if you want to go deeper in generic DRL
- ❑ 16-881 Seminar: Deep Reinforcement Learning for Robotics (Deepak Pathak)
- ❑ 16-884 Deep Learning for Robotics (David Held)
 - Pure paper discussions or mix of lectures and paper discussions
 - Not basic, assume prior background
 - Recommend if you want to learn latest robot learning research
- ❑ 15-889 Special Topics in AI: Adv. Topics in Reinforcement Learning and Decision Making (Aviral Kumar)
 - Advanced RL algorithms and concepts, not necessarily for robotics
 - Recommend if you want to dive deeper in RL

Relation to Other CMU Courses

❑ 16-745 Optimal Control and Reinforcement Learning (Zachary Manchester)

- Focus on optimization-based control and model-based RL
- Recommend if you want to know more about lecture 13-14

10/08	Lecture 13: Optimal Control and Planning Part 1	[Murray's Notes][iLQR] [DDP][SCP]		 Model-Based RL
10/10	Lecture 14: Optimal Control and Planning Part 2	[PETS][Neural-Control Family][MPPI][PILCO] [MBPO]	Project Proposal Due	 Model-Based RL

❑ 16-899 Adaptive Control and Reinforcement Learning (Changliu Liu)

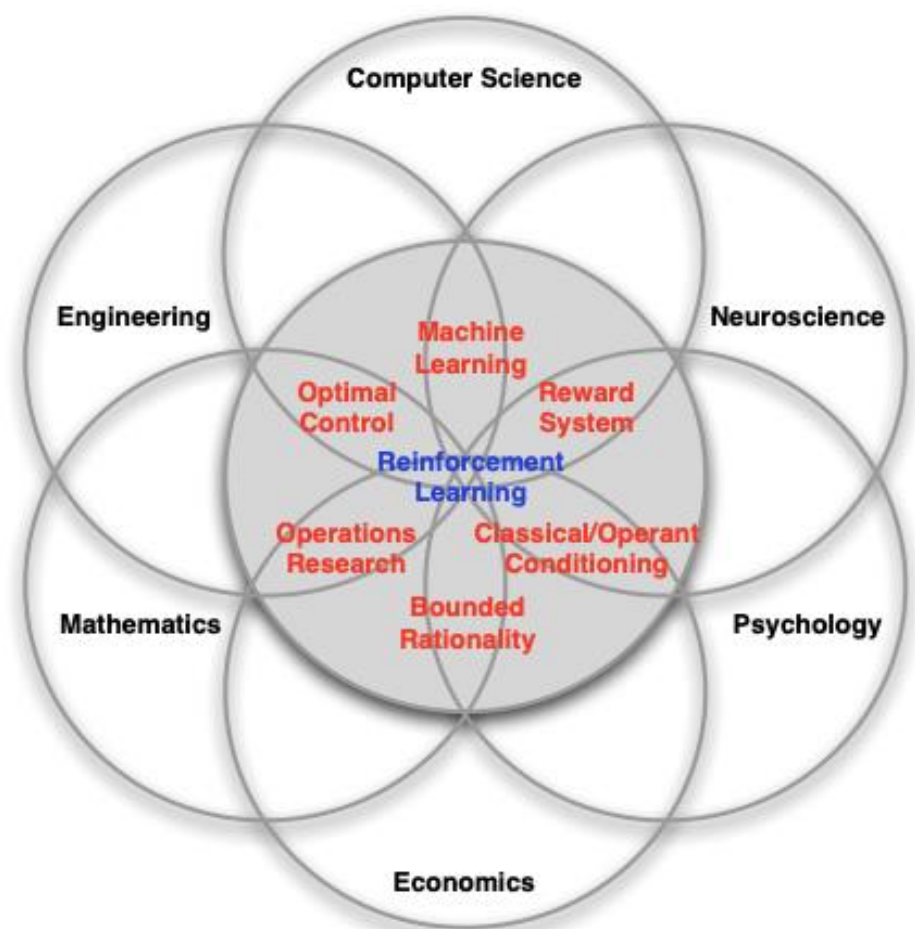
- Advanced course focusing on control theory, safe control, Lyapunov stability, etc
- Recommend if you want to know more about lecture 12 and safe RL

Today's Agenda

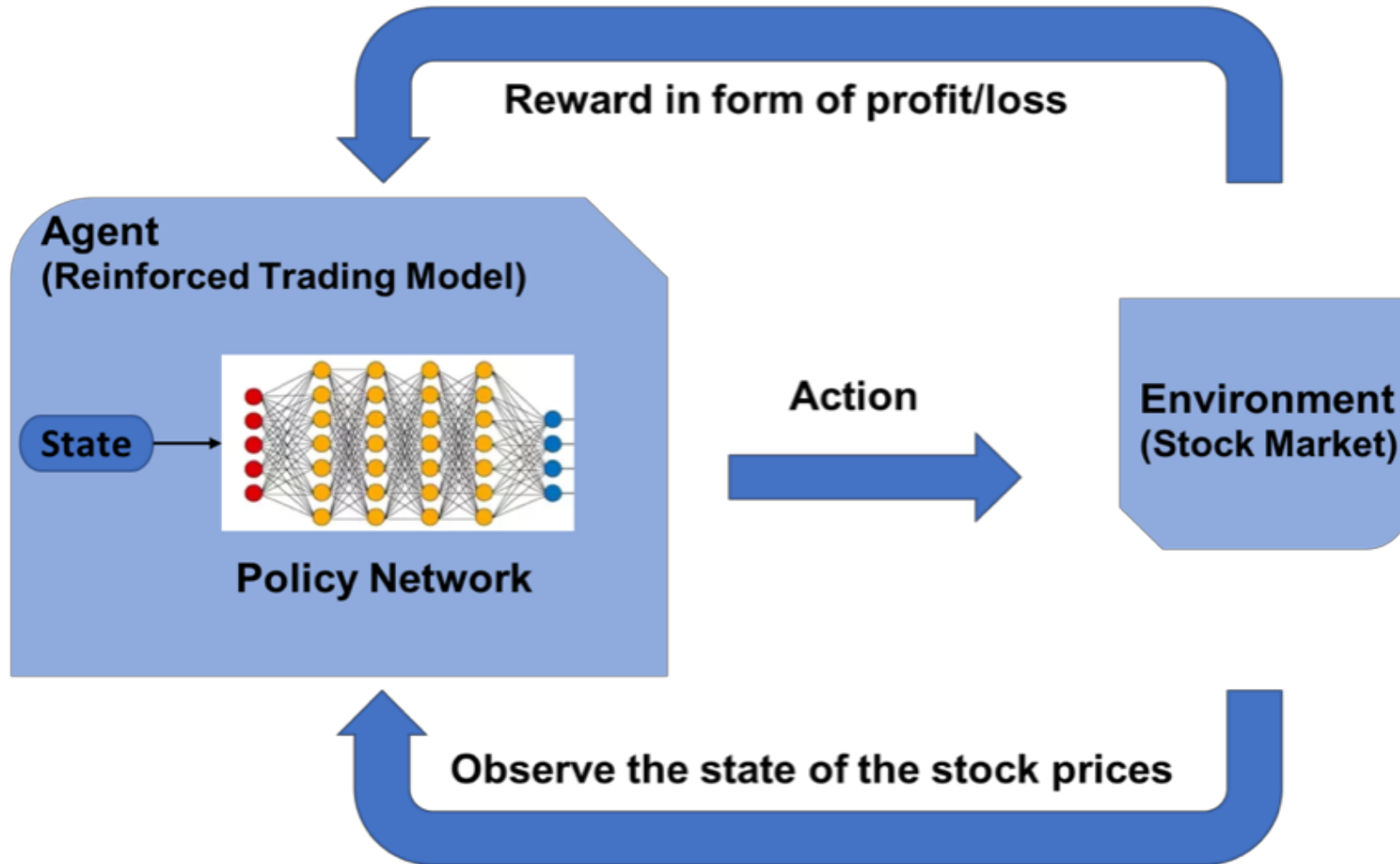
- ☐ What is robot learning and what is its uniqueness?
- ☐ What is the goal of robot learning?
- ☐ Where are we today? How far are we from the goal?
- ☐ Course structure and course objective
- ☐ Relation to other CMU courses
- ☒ Applications of sequential decision making in other domains

Other Applications of Sequential Decision Making

- ❑ This course focuses on robotics and embodied intelligence
- ❑ But the methods it covers apply to other sequential decision making domain!



RL for Trading and Finance



RL for Cluster Management

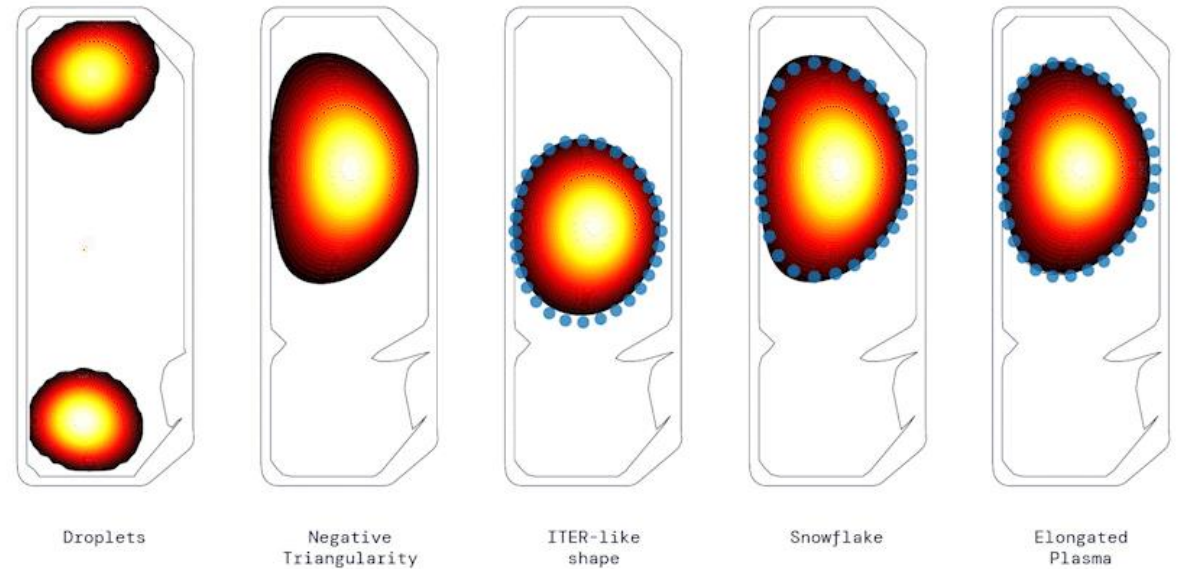
- ❑ RL to figure out where to best run neural network models at any given time as model ensembles are running on a heterogeneous distributed environment with a mixture of hardware devices such as CPUs and GPUs

Device Placement Optimization with Reinforcement Learning

Azalia Mirhoseini^{*12} Hieu Pham^{*12} Quoc V. Le¹ Benoit Steiner¹ Rasmus Larsen¹ Yuefeng Zhou¹
Naveen Kumar³ Mohammad Norouzi¹ Samy Bengio¹ Jeff Dean¹

RL for Nuclear Fusion through Plasma Control

❑ <https://deepmind.google/discover/blog/accelerating-fusion-science-through-learned-plasma-control/>



RL for Nuclear Fusion through Plasma Control

□ <https://www.cmu.edu/news/stories/archives/2022/september/exploring-reinforcement-learning-to-control-nuclear-fusion-reactions>

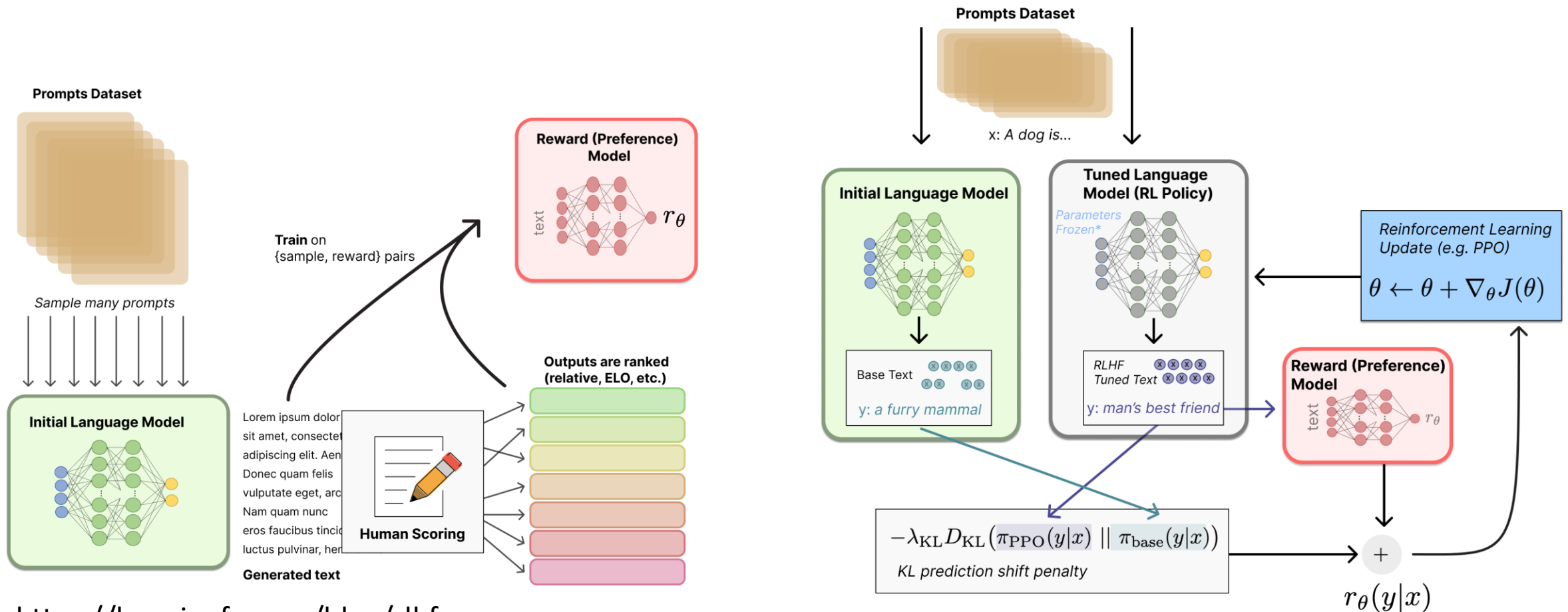
Offline Model-Based Reinforcement Learning for Tokamak Control

Ian Char, Joseph Abbate, Laszlo Bardoczi, Mark Boyer, Youngseog Chung, Rory Conlin, Keith Erickson, Viraj Mehta, Nathan Richner, Egemen Kolemen, Jeff Schneider Proceedings of The 5th Annual Learning for Dynamics and Control Conference, PMLR 211:1357-1372, 2023.



RL for Training LLMs

- ❑ Reinforcement learning from human feedback (RLHF) for training LLMs
- ❑ Particularly important for alignment



Thank You!